LEDAW Manual

LED Analysis Wizard

A Python Program with GUI for Automating ORCA Input Generation and Output Processing for Local Energy Decomposition Analysis

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INTRODUCTION

CCSD(T)[1] is widely regarded as the "gold standard" in quantum chemistry; however, its computational cost scales steeply with system size, limiting its applicability to small molecules. Its "Domain-based Local Pair Natural Orbital" variant, DLPNO-CCSD(T),[2–6] exploits the rapid decay of the electron correlation with the inter-electronic distance, achieving near-linear scaling with respect to system size. Consequently, it can be applied to sizable biomolecules and complex materials, such as entire proteins, with accuracy comparable to the canonical CCSD(T). The Hartree-Fock (HF) plus London Dispersion (LD) scheme (*i.e.*, HFLD)[7,8] introduces physically relevant assumptions to DLPNO-CCSD(T), further improving its efficiency in analyzing noncovalent interactions (NCIs). HFLD consistently provides accuracy between those of CCSD and CCSD(T) on challenging closed-shell and open-shell benchmark sets for NCIs.[7,8] As a result, HFLD has been extensively used in large-scale computational studies involving proteins,[7] solute-solvent interactions,[8] DNA,[9] and crystal structures.[10–12]

In DLPNO methods, truncation of the PNO space significantly enhances computational efficiency. With "TightPNO" cutoffs, DLPNO-CCSD(T) already achieves an accuracy of 0.5 kcal/mol on the entire GMTKN55 benchmark superset.[13] However, for complex molecular interactions and benchmark calculations, even tighter settings than TightPNO may be required—limiting the method's applicability. Instead, the efficient two-point "Complete PNO Space (CPS)" extrapolation scheme can be used to obtain results at the complete PNO space limit.[14]

Although DLPNO error is a size-extensive quantity that increases with molecular system size, CPS extrapolation eliminates its size dependence, enabling accurate results even for large systems.[15] In particular, CPS(6/7) extrapolation of DLPNO-CCSD(T)/TightPNO energies —utilizing results with $T_{CutPNO} = 10^{-6}$ and $T_{CutPNO} = 10^{-7}$— achieves sub-kJ/mol (~0.24 kcal/mol) accuracy for challenging subsets of GMTKN55, relative to canonical CCSD(T) with the corresponding basis set.[14]

It is well known that correlation methods exhibit much slower basis set convergence compared to mean-field approaches, such as density functional theory (DFT). Since computations with sufficiently large basis sets are often prohibitively expensive for correlated methods, two-point "Complete Basis Set (CBS)" extrapolation schemes[16,17] are extensively employed to estimate results at the complete basis set limit. Hence, it is worth emphasizing that achieving benchmark accuracy with DLPNO-CCSD(T) requires both CBS and CPS extrapolations.

To facilitate the interpretation of energies computed by highly accurate and efficient DLPNO-based methods, the Local Energy Decomposition (LED) scheme, which breaks down these energies into several physically meaningful terms (electrostatic, exchange, LD, electronic preparation, etc.), was developed.[18] Its original formulation, designed to analyze interactions between pairs of fragments in closed-shell[18] and open-shell[19] systems (with inherent ability for multi-fragment analyses) across both weak and strong interaction regimes, has already found widespread applications in diverse fields, significantly enhancing the understanding of complex molecular interactions.[20–28] Over the years, several extensions, offering deeper insights into the nature of complex chemical interactions, were introduced to the LED scheme. For example, the frozen-state LED scheme[29] allowed for the extraction of permanent and inductive (orbital relaxation) components of decomposed LED terms. While LD between pairs of fragments is calculated in the original LED formalism at the DLPNO-CCSD level, it has since become possible[26,30] to estimate the triples contribution to LD, allowing for a better assessment[20] of the role played by LD in chemical reactions. Decomposing LD computed with the LED scheme to atomic contributions is also possible through the "Atomic Decomposition of London Dispersion energy (ADLD)" method.[31] An LED scheme that allows fragment definitions through covalent bonds (CovaLED)[32] is also available in ORCA 6.1+.

Applications of the standard LED scheme to large molecular systems consisting of more than two fragments, such as molecular clusters,[33] biomolecular assemblies,[9] protein-ligand interactions,[34,35] and molecular crystals,[10] have demonstrated its versatility and potential to drive breakthroughs across various domains of chemical research, including materials science, catalysis, and drug discovery.[36] Although LED applications in its standard formalism are useful in revealing the trends in the interaction strengths of fragment pairs, they cannot provide exact interaction energies of fragment pairs within the entire system comparable to those of the isolated fragment pairs.

This limitation prompted the development of the fragment-pairwise (fp)-LED scheme,[30] which achieve fully fragment-pairwise exact energy decomposition by further breaking down specific energy terms. In particular, the intra-fragment component of the interaction energy, known as electronic preparation, arises from the perturbation of a given fragment's energy by all other fragments in the system, and thus it is inherently cumulative. While the standard LED scheme treats this term as a single quantity per fragment, the fp-LED scheme further dissects it into fragment-pairwise components, explicitly quantifying how much each fragment's energy is perturbed by every other fragment in the system.[30] Additionally, the fp-LED scheme distributes the solvent contributions, such as dielectric energy and cavity-dispersion-solvent-structure (CDS), representing direct solute-solvent interactions in implicit solvation calculations, into fragment-pairwise contributions.[30]

The LED scheme decomposes the interaction energy computed within a supramolecular approach into intra-fragment contributions —further distributed in the fp-LED approach—

and inter-fragment contributions. The sum of these terms fully recovers the original interaction energy. In multi-fragment systems, these terms inherently incorporate environmental effects, often referred to as the "many-body" or "cooperativity" effects. Hereafter, we will refer to the LED analysis of the entire chemical system as "N-body LED" and that of the isolated fragment pairs as "two-body LED". By subtracting two-body values from their N-body counterparts, one can directly quantify cooperativity contribution to each energy term.[30,33] This straightforward approach enables the LED scheme to provide a robust and quantitative framework for investigating cooperativity effects in molecular interactions.

Since the LED scheme decomposes interaction energy computed within supramolecular approach, LED analyses require ORCA[37–41] output files not only for the supersystem but also for the corresponding subsystems. Additionally, CBS and CPS extrapolations necessitate ORCA output files for each computational setting. Various extensions to the LED scheme also require additional computations using data extracted from these outputs. As a result, extracting and processing LED data from these text-based ORCA outputs demands strong text-mining, data analysis, and data visualization skills. Therefore, even for experienced users, conducting a complete LED analysis of a multi-fragment system can be highly time-consuming, often taking several hours to several days. Furthermore, the initial LED input preparation for ORCA can be an overwhelming task for multi-fragment systems.

Given these challenges, we frequently received inquiries regarding the application and execution of LED analyses. These inquiries prompted us to write the "LED Analysis Wizard (LEDAW)" program.[42] LEDAW is a comprehensive program that automates the entire LED analysis workflow, from ORCA input file preparation to final data visualization. It consists of two main components: `ledip` for pre-processing, which handles the complex task of automatic fragmentation and ORCA input file generation, and `ledaw` for post-processing, which extracts data from ORCA output files, calculates each LED interaction term within standard and fp-LED schemes, compiles the results into interaction energy matrices, and generates LED interaction energy heat maps. The program fully automates all kinds of LED analyses, including CBS and CPS extrapolations, and features an intuitive graphical user interface (GUI) that requires users to simply specify the paths to the individual ORCA output files. Once the required paths are provided, `ledaw` delivers ready-to-use LED analysis results typically within a minute, even if all kinds of LED analyses and extrapolation schemes are requested in a single run.

In the following, we will first provide a brief overview of the LED scheme for different types of analyses. Next, we will describe the LEDAW software package. Then, we will demonstrate how to prepare ORCA input files for LED computations across various molecular systems and interaction types and how to automate this task with `ledip`. Finally, we will show how to use `ledaw` with its GUI (LEDAW-GUI) to obtain LED analysis results of these computations.

2

A BRIEF OVERVIEW OF LED

2.1. Supersystem, Subsystem, and Fragment Definitions

The LED scheme enables dissecting the interaction energy of the entire molecular system (referred to as the "supersystem") —comprising two or more complex chemical entities (referred to as the "subsystem"), each potentially composed of multiple fragments— into contributions from each fragment pairs. In the following, we will frequently use the supersystem, subsystem, and fragment terms. Therefore, it is essential to clarify their distinctions (see Figure 2.1):

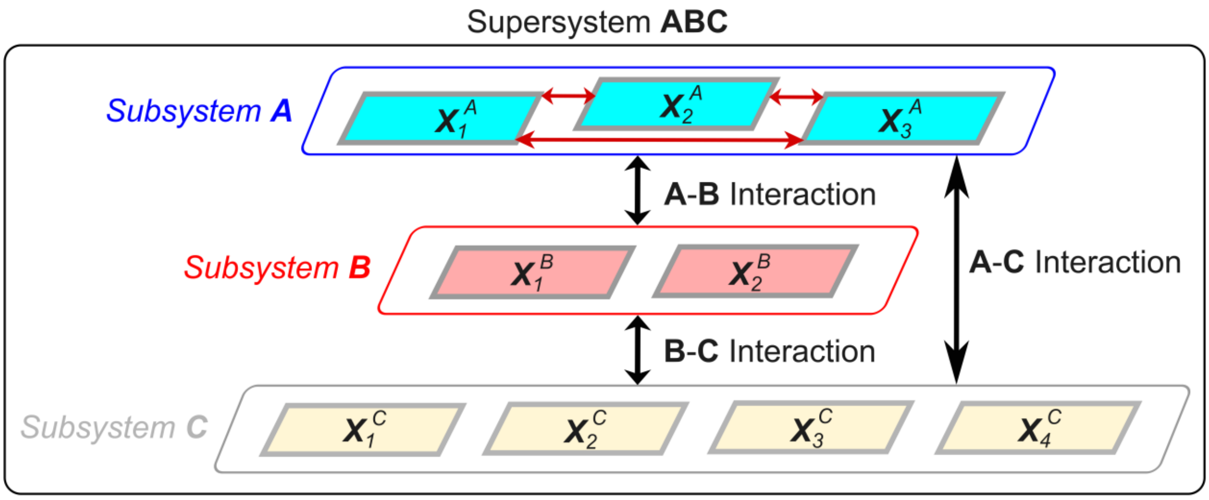


Figure 2.1. Schematic representation of a supramolecular complex ABC (referred to as the "supersystem" or "entire system"), comprising three subsystems: A, B, and C. Each subsystem contains multiple fragments: Subsystem A includes X_i^A (i = 1, 2, 3) fragments; Subsystem B includes X_i^B (i = 1, 2) fragments; Subsystem C includes X_i^C (i = 1, 2, 3, 4) fragments. Interactions between subsystems are represented by double-ended black arrows labeled A-B, B-C, and A-C. They are the sum of all fragment-pairwise interactions across the corresponding subsystem pairs, such as X_1^A- X_1^B, X_1^A- X_2^B, X_1^A- X_4^C, etc. For simplicity, the intra-subsystem interactions (*e.g.*, X_1^A- X_2^A) are shown only within subsystem A, depicted by double-ended red arrows.

As an example, when investigating inter-strand stability of an RNA or DNA triplex, the supersystem is the complete triplex under consideration; the three strands are subsystems; and the nucleotide bases on the strands are fragments. Similarly, in studying interactions between the layers of a supramolecular complex or a molecular crystal, the entire supramolecular assembly under consideration is the supersystem; the layers are subsystems; and each individual molecular unit is a fragment. If cluster formation energy of a water hexamer is under consideration, then each water molecule serves as both a fragment and a subsystem.

2.2. N-Body LED of a Supersystem

As this is a practical guide to the LED scheme, in the following we will provide only the necessary equations to understand and interpret the LED results. For the underlying theory, readers are referred to the original LED publications.

2.2.1. Basic Formulation of N-Body LED: Standard LED vs. fp-LED

To perform an LED computation, the supersystem must first be divided into two or more fragments. If the user does not specify fragment information in the ORCA input file of an LED computation for a noncovalently interacting system, ORCA automatically assigns each noncovalently interacting molecule as a separate fragment. The LED scheme then decomposes the total energy of the supersystem E into intra-fragment $E^{(X)}$ and inter-fragment $E^{(X,Y)}$ components, where X and Y correspond to different fragments.

$$E = \sum_{X} E^{(X)} + \sum_{X>Y} E^{(X,Y)} \tag{1}$$

Eq. (1) does not account for subsystem information. Therefore, to demonstrate how to compute interaction energy in a multi-fragment system, it is beneficial to relabel fragments by indicating both the subsystem they belong to and their index within that subsystem, as illustrated in Figure 2.1:

$$E = \sum_{K} \left[\sum_{i} E^{(X_i^K)} + \sum_{i>j} E^{(X_i^K, X_j^K)} \right] + \sum_{K>L} \sum_{i \geq j} E^{(X_i^K, X_j^L)} \tag{2}$$

Here X_i^K and X_j^L represent two fragments, where K and L label subsystems, while i and j (taking values of 1, 2, 3, …) label fragments within the corresponding subsystems. The first part in Eq. (2) that includes two summations inside a parenthesis represents the sum of subsystem energies within the supersystem, incorporating both the fragment energies of a given subsystem, $E^{(X_i^K)}$, and the interaction energies of fragment pairs within the same subsystem, $E^{(X_i^K, X_j^K)}$. The second part accounts for the sum of interaction energies between fragments located in different subsystems, $E^{(X_i^K, X_j^L)}$.

If a subsystem is also a multi-fragment system, then it is also necessary to perform LED analysis on these subsystems. Then, the sum of isolated subsystem energies becomes:

$$E_{isol} = \sum_{K} \left[\sum_{i} \tilde{E}^{(X_i^K)} + \sum_{i>j} \tilde{E}^{(X_i^K, X_j^K)} \right] \tag{3}$$

Note that the form of Eq. (3) is identical to the first part of Eq. (2), except that while this quantity is obtained in Eq. (2) on the supersystem, it is computed in Eq. (3) on the isolated subsystems. To highlight this distinction, subsystem energy components are denoted as $\tilde{E}$ rather than as E. It is worth noting that, for a one-fragment subsystem, the $E^{(X_i^K, X_j^K)}$and $\tilde{E}^{(X_i^K, X_j^K)}$ terms in Eqs. (2) and 3 are simply zero.

The interaction energy of subsystems ΔE_{int} can be computed by subtracting E_{isol} from E:

$$\Delta E_{int} = \sum_K \left[\sum_i (E^{(X_i^K)} - \tilde{E}^{(X_i^K)}) + \sum_{i>j} (E^{(X_i^K, X_j^K)} - \tilde{E}^{(X_i^K, X_j^K)}) \right] + \sum_{K>L} \sum_{i \geq j} E^{(X_i^K, X_j^L)} \tag{4}$$

The first subtraction term in Eq. 4 corresponds to the perturbation of an individual fragment energy by the presence of other fragments and called as electronic preparation energy $\Delta E_{el-prep}^{(X_i^K)}$. The second subtraction term represents the perturbation of interaction energy of a fragment pair within the same subsystem by the rest of the supersystem, denoted as $\Delta E^{(X_i^K, X_j^K)}$. By inserting their abbreviations in Eq. (4), we obtain:

$$\Delta E_{int} = \sum_K \left[\sum_i \Delta E_{el-prep}^{(X_i^K)} + \sum_{i>j} \Delta E^{(X_i^K, X_j^K)} \right] + \sum_{K>L} \sum_{i \geq j} E^{(X_i^K, X_j^L)} \tag{5}$$

Eq. (5) is the final form of ΔE_{int} within the standard LED scheme. However, to simplify the remainder of this section, we revert to the original fragment labeling (X, Y) introduced at the beginning of this section and rewrite Eq. (5) in a more compact form as

$$\Delta E_{int} = \sum_X \Delta E_{el-prep}^X + \sum_{X>Y} \varepsilon^{(X,Y)} \tag{6}$$

Here, $\varepsilon^{(X,Y)} \equiv \Delta E^{(X,Y)}$ when X and Y belong to the same subsystem, and $\varepsilon^{(X,Y)} \equiv E^{(X,Y)}$ when X and Y are on different subsystems.

$\Delta E_{el-prep}^X$ arises from the perturbation of fragment X by all the other fragments, making it a cumulative quantity. The fp-LED scheme decouples this cumulative energy by determining the magnitude of deformation of each given fragment energy by each of the other fragments. In two-fragment systems, a strong linear correlation exists between inter-fragment interaction $\varepsilon^{(X,Y)}$ and electronic preparation energy, *i.e.*, $\Delta E_{el-prep}^{XY} = \Delta E_{el-prep}^X + \Delta E_{el-prep}^Y$.[30] In other words, the larger the inter-fragment X-Y interaction, the larger the perturbation of fragment X and Y by each other. Hence, in multi-fragment systems, $\Delta E_{el-prep}^{XY}$ can be determined by scaling the cumulative $\Delta E_{el-prep}^X$ and $\Delta E_{el-prep}^Y$ quantities with the corresponding $\varepsilon^{(X,Y)}$ values:[30]

$$\Delta E_{el-prep}^{XY} = \omega_X^{(X,Y)} \Delta E_{el-prep}^X + \omega_Y^{(X,Y)} \Delta E_{el-prep}^Y \tag{7}$$

where the weight factors are as:

$$\omega_X^{(X,Y)} = \frac{|\varepsilon^{(X,Y)}|}{\sum_{K \neq X} |\varepsilon^{(X,K)}|} \quad ; \quad \omega_Y^{(X,Y)} = \frac{|\varepsilon^{(X,Y)}|}{\sum_{K \neq Y} |\varepsilon^{(Y,K)}|} \tag{8}$$

The denominators in Eq. (8) ensure that

$$\sum_X \Delta E_{el-prep}^X = \sum_{X,Y} \Delta E_{el-prep}^{X,Y} \tag{9}$$

Inserting Eq. (9) into Eq. (6), a fully pairwise decomposition of ΔE_{int} is achieved in the framework of the fp-LED scheme:

$$\Delta E_{int} = \sum_{X>Y} \Delta E_{el-prep}^{XY} + \sum_{X>Y} \varepsilon^{(X,Y)} = \sum_{X>Y} \Delta E_{int}^{XY} \tag{10}$$

In the original fp-LED formulation,[30] the ω weight factors (see Eq. (8)) were defined without absolute values, as $\varepsilon^{(X,Y)}$ is typically negative for NCIs, resulting in positive ω factors. However, $\varepsilon^{(X,Y)}$ can sometimes be positive, such as when the fragments are connected through covalent bonds, leading to artificially negative ω factors if absolute values are omitted. Therefore, to ensure general applicability, the LEDAW implementation of fp-LED incorporates absolute values in Eq. (8).

2.2.2. Comparison of Standard LED and fp-LED Interaction Energy Heat Maps

In multi-fragment systems, LED results are most effectively illustrated and analyzed using interaction energy matrices or their heat map representations, which provide an immediate and intuitive visualization of pairwise interaction strengths. As an example, standard LED (based on Eq. (6)) and fp-LED (based on Eq. (10)) heat maps for the inter-strand interaction energy of a DNA duplex segment are shown in Figure 2.2.

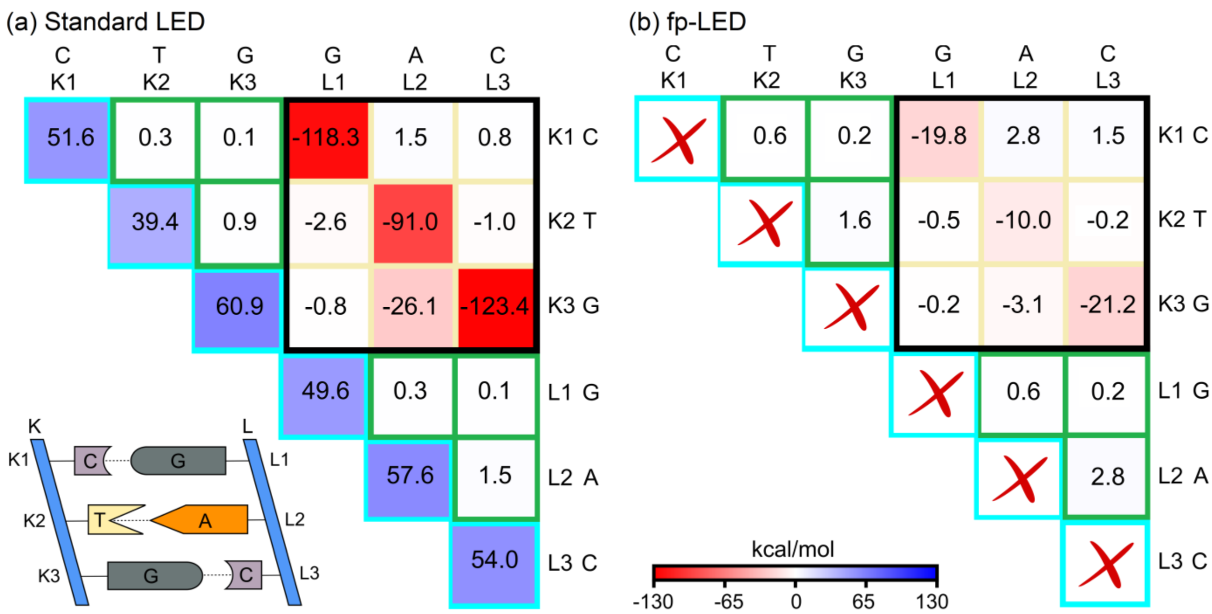


Figure 2.2. (a) Standard LED and (b) fp-LED heat maps for the BSSE-corrected total interaction energy between strands K and L of a three-nucleobase-long DNA duplex. Calculations were performed at the DLPNO-CCSD(T_1)/NormalPNO*/def2-TZVP(-f) level, treating the water environment implicitly with the CPCM model. The CPCM dielectric term was distributed as described in Section 2.2.3. The supersystem corresponds to the entire supramolecular complex KL, with backbones omitted. As illustrated in the schematic DNA ladder at the bottom of panel (a), both subsystems K and L consist of three fragments: Subsystem K includes K1 (C), K2 (T), and K3 (G) while subsystem L includes L1 (G), L2 (A), and L3 (C). The heat maps generated by LEDAW were slightly modified by coloring the borders of each cell according to the corresponding LED energy category in Eqs. (6) and (10) and labeling the fragments with their names instead of numerical indices. Coordinates were taken from Ref. 9. In the heat maps, red represents attractive interactions, whereas blue represents repulsive interaction

The diagonal elements in the standard LED heat map (bluish elements enclosed by turquoise lines in Figure 2.2a) represent the $\Delta E_X^{el-prep}$ terms in Eq. (6). These terms correspond to the energy cost associated with the deformation of the electronic structure of each fragment upon duplex formation. The nondiagonal elements enclosed by green lines represent the $\varepsilon^{(X,Y)} \equiv \Delta E^{(X,Y)}$ terms, which account for the deformation energy of fragment pairs located within the same strand (upper triangle for strand K; lower triangle for strand L). Since these terms are defined for fragment pairs, they do not exist for subsystems consisted of a single fragment.

The submatrix enclosed by a black box contains the $\varepsilon^{(X,Y)} \equiv E^{(X,Y)}$ terms, which represent the genuine inter-strand interaction energies between fragment pairs where the two fragments are located on different strands.

As shown in Figure 2.2a, the standard LED interaction energy map contains very large positive values ($\Delta E_X^{el-prep}$) and large negative values ($E^{(X,Y)}$). However, upon redistributing the repulsive $\Delta E_X^{el-prep}$ terms to the pairwise interactions within the fp-LED scheme, the resulting fp-LED heat map (Figure 2.2b) directly provides individual interaction energy strengths. These values are comparable to those of isolated dimers while inherently including many-body effects.

The diagonal elements of the submatrix enclosed by the black box are the strongest pairwise energy terms in this DNA example, highlighting that base-pairing (hydrogen bonding between nucleobases) is the primary contributor to duplex formation. Although significantly smaller, the stacking interactions (see the elements just above and below these diagonal entries) are still noticeable.

Finally, it is worth noting that although fp-LED terms offer more intuitive and directly comparable values relative to isolated systems, both standard LED and fp-LED approaches yield consistent conclusions in trend analyses.

2.2.3. Further Decomposition of N-Body LED Terms

For an in-depth analysis of chemical interactions, the terms in Eq. (6) (corresponding to the standard LED scheme) and in Eq. (10) (corresponding to the fp-LED scheme) can be further decomposed into some physically meaningful components. Each term in these equations consists of a reference component (typically RHF for closed-shell systems and QRO for open-shell systems) and a correlation component:

$$\Delta E_{int} = \Delta E_{int}^{ref} + \Delta E_{int}^{C} \tag{11}$$

where

$$\Delta E_{int}^{ref} = \sum_{X} \Delta E_{el-prep}^{X,ref} + \sum_{X>Y} \varepsilon^{(X,Y),ref} = \sum_{X>Y} \Delta E_{el-prep}^{XY,ref} + \sum_{X>Y} \varepsilon^{(X,Y),ref} = \sum_{X>Y} \Delta E_{int}^{XY,ref} \tag{12}$$

$$\Delta E_{int}^{C} = \sum_{X} \Delta E_{el-prep}^{X,C} + \sum_{X>Y} \varepsilon^{(X,Y),C} = \sum_{X>Y} \Delta E_{el-prep}^{XY,C} + \sum_{X>Y} \varepsilon^{(X,Y),C} = \sum_{X>Y} \Delta E_{int}^{XY,C} \tag{13}$$

To compute $\Delta E_{el-prep}^{XY,ref}$, the $\varepsilon^{(X,Y),ref}$ and $\Delta E_{el-prep}^{X,ref}$ terms must be used in Eqs. (7) and (8). Since HFLD contains only the reference electronic preparation, $\Delta E_{el-prep}^{XY}$ of HFLD corresponds to $\Delta E_{el-prep}^{XY,ref}$. For DLPNO-CCSD(T) and DLPNO-CCSD, $\Delta E_{el-prep}^{XY,C}$ can be evaluated through two primary approaches:

1) Direct computation from correlation components, *i.e.*, using $\varepsilon^{(X,Y),C}$ and $\Delta E_{el-prep}^{X,C}$ in Eqs. (7) and (8);

2) Subtracting total and reference components. This approach is based on once using $\varepsilon^{(X,Y)}$ and $\Delta E_{el-prep}^{X}$ and once using $\varepsilon^{(X,Y),ref}$ and $\Delta E_{el-prep}^{X,ref}$ in Eqs. (7) and (8), then subtracting the resulting $\Delta E_{el-prep}^{XY,ref}$ from $\Delta E_{el-prep}^{XY}$.

These two approaches yield negligible differences.[30] Therefore, to maintain consistency with the original fp-LED formalism, we adopt the second approach in the LEDAW implementation.

The $\varepsilon^{(X,Y),ref}$ component can be decomposed into electrostatic and two-electron exchange components:

$$\varepsilon^{(X,Y),ref} = \varepsilon_{elstat}^{(X,Y)} + \varepsilon_{exch}^{(X,Y)} \tag{14}$$

Among many ways of grouping correlation terms within $\varepsilon^{(X,Y),C}$, the most intuitive way is to separate it into dispersion (LD) and non-dispersion components:

$$\varepsilon^{(X,Y),C} = \varepsilon_{disp}^{(X,Y)} + \varepsilon_{no-disp}^{(X,Y)} \tag{15}$$

Together with the correlation electronic preparation $\Delta E_{el-prep}^{XY,C}$, the non-dispersion term $\varepsilon_{no-disp}^{(X,Y)}$ serves as a correction to the electrostatic interaction calculated at the HF level due to the characteristic overestimation of dipole moments at the HF level. Therefore, these two terms are sometimes combined as $\Delta\varepsilon_{no-disp}^{(X,Y)}$. At the DLPNO-CCSD and HFLD levels, $\varepsilon_{disp}^{(X,Y)}$ is the sum of strong pairs $\varepsilon_{disp}^{(X,Y),C-SP}$and weak pairs $\varepsilon_{disp}^{(X,Y),C-WP}$ dispersion terms. At the DLPNO-CCSD(T) level, $\varepsilon_{disp}^{(X,Y)}$ additionally includes triples dispersion $\varepsilon_{disp}^{(X,Y),C-(T)}$, calculated using Eq. (16):[26,30]

$$\varepsilon_{disp}^{(X,Y),C-(T)} = \gamma \cdot \varepsilon_{disp}^{(X,Y),C-SP} \quad ; \quad \gamma = \frac{\varepsilon^{(X,Y),C-(T)}}{\varepsilon^{(X,Y),C-SP}} \tag{16}$$

In implicit solvation calculations, the electronic energy includes an additional term known as dielectric contribution, ΔE_{diel}. Due to strong linear correlation between ΔE_{int}^{XY} and $\Delta E_{diel}^{(X,Y)}$ in two-fragment systems,[30] this global quantity, which represents the direct total interaction between the supersystem and implicit solvent molecules, can be decomposed into pairwise terms within the fp-LED scheme as:

$$\Delta E_{diel}^{(X,Y)} = \alpha \cdot \Delta E_{int}^{XY} \quad ; \quad \alpha = \frac{\Delta E_{diel}}{\sum_{X>Y} \Delta E_{int}^{XY}} \tag{17}$$

When $\Delta E_{diel}^{(X,Y)}$ terms are then added to the corresponding ΔE_{int}^{XY} terms –which are devoid of dielectric contribution–, total interaction energy of the XY fragment pair within a complex molecular system is obtained.

This dielectric distribution scheme can also be applied in the standard LED framework by scaling all components of the interaction energy with the same constant α:

$$\Delta E_{diel}^{(X,Y)} = \alpha \cdot \varepsilon^{(X,Y)} \quad ; \quad \Delta E_{diel}^{X} = \alpha \cdot \Delta E_{el-prep}^{X} \quad ; \quad \alpha = \frac{\Delta E_{diel}}{\sum_X \Delta E_{el-prep}^{X} + \sum_{X>Y} \varepsilon^{(X,Y)}} \tag{18}$$

The resulting $\Delta E_{diel}^{(X,Y)}$ and ΔE_{diel}^{X}terms are then added to the corresponding $\varepsilon^{(X,Y)}$ and $\Delta E_{el-prep}^{X}$ terms. If the fp-LED electronic preparation distribution scheme is applied after this addition, the same pairwise total interaction energies are obtained as those found when the dielectric contribution is distributed after obtaining fp-LED electronic preparation. Hence, the order in which electronic preparation and dielectric energy distribution schemes are applied does not affect the final results.

In Eqs. (17) and (18), the denominator of the scaling factor α is the sum of all terms in the interaction energy matrix, corresponding to the total interaction energy in the absence of dielectric contribution. Thus, the coefficient α is actually the same in both standard and fp-LED schemes. The denominator ensures that

$$\sum_{X>Y} \Delta E_{diel}^{(X,Y)} = \Delta E_{diel} \quad (19)$$

for the fp-LED scheme, and

$$\sum_{X} \Delta E_{diel}^{X} + \sum_{X>Y} \Delta E_{diel}^{(X,Y)} = \Delta E_{diel} \quad (20)$$

for the standard LED scheme.

Coupled-cluster solvation schemes may include a typically very small correlation correction to the reference-level CPCM dielectric term. Additionally, implicit solvation schemes such as Solvation Model Density (SMD) also incorporate a nonelectrostatic solute-solvent interaction term, known as the cavity-dispersion-solvent-structure (CDS). In LEDAW, these solvation terms are distributed using the α factor described in Eqs. (17) and (18). For extrapolations to the CBS and/or CPS limit, LEDAW treats all implicit solvation terms as the reference contribution."

2.3. Two-Body LED and Cooperativity

In systems where multiple fragments interact, the interaction energy between each fragment pair is influenced by the presence of surrounding fragments. This environmental influence is often referred to as the many-body or cooperativity contribution. Beyond the N-body LED analysis, evaluating the interaction energy within the two-body approximation of the many-body expansion (MBE), along with its LED components, offers a framework to dissect these many-body effects. This approach entails additional yet computationally inexpensive LED calculations on each isolated fragment pair, comprising fragments from different subsystems.

While N-body terms inherently incorporate the influence of the surrounding chemical environment, two-body terms lack this effect as they are obtained on isolated fragment pairs. Therefore, by subtracting a two-body LED term from the corresponding N-body LED term, one can isolate the contribution of the chemical environment to that interaction term. This strategy enables quantification of not only the exact overall cooperative effect but also its physically meaningful components.[30,33]

Within the LED scheme, the two-body interaction energy $\Delta E_{int}^{(2)}$ is expressed as:

$$\Delta E_{int}^{(2)} = \sum_{X>Y} \left[\Delta E_{el-prep-XY}^{X,(2)} + \Delta E_{el-prep-XY}^{Y,(2)} + \varepsilon^{(X,Y),(2)}\right] \quad (21)$$

Here, the superscript (2) indicates the two-body approximation and thus results on isolated fragment pairs. In the two-body analysis, only fragments X and Y from different subsystems (K and L) are considered. Hence, the $E^{(X,Y),(2)}$ components are zero because X and Y in these terms refer to the same subsystem (X, $Y \in K$). This simplifies the fragment-pairwise two-body interaction term to $\varepsilon^{(X,Y),(2)} = E^{(X,Y),(2)}$, with $X \in K$ and $Y \in L$. $\Delta E_{el-prep-XY}^{X,(2)}$ and $\Delta E_{el-prep-XY}^{Y,(2)}$ are electronic preparation energy components of an isolated fragment pair XY. In contrast, in N-body analysis, electronic preparation energy $\Delta E_{el-prep}^{X}$ is a cumulative term, encompassing contributions from all relevant fragment pairs involving fragment X. Hence, to align standard N-body and two-body analyses, $\Delta E_{el-prep-XY}^{X,(2)}$ from each fragment pair must be aggregated as:[33]

$$\Delta E_{el-prep}^{X,(2)} = \sum_{Y} \Delta E_{el-prep-XY}^{X,(2)} \quad (22)$$

This leads:

$$\Delta E_{int}^{(2)} = \sum_{X} \Delta E_{el-prep}^{X,(2)} + \sum_{X>Y} \varepsilon^{(X,Y),(2)} \tag{23}$$

The cooperativity contribution ΔE_{int}^{coop} in the standard LED framework is then obtained by subtracting Eqs. (6) and (23) side by side:

$$\Delta E_{int}^{coop} = \Delta E_{int} - \Delta E_{int}^{(2)} = \sum_{X} \left[\Delta E_{el-prep}^{X} - \Delta E_{el-prep}^{X,(2)}\right] + \sum_{X>Y} \left[\varepsilon^{(X,Y)} - \varepsilon^{(X,Y),(2)}\right]$$

$$= \sum_{X} \Delta E_{el-prep}^{X,coop} + \sum_{X>Y} \varepsilon^{(X,Y),coop} \tag{24}$$

In the fp-LED framework, $\Delta E_{el-prep}^{XY,(2)}$ can be expressed as:

$$\Delta E_{el-prep}^{XY,(2)} = \Delta E_{el-prep-XY}^{X,(2)} + \Delta E_{el-prep-XY}^{Y,(2)} \tag{25}$$

This expression is exact because two-body analysis is performed on isolated fragment pairs. Consequently, the deformation energies $\Delta E_{el-prep-XY}^{X,(2)}$ and $\Delta E_{el-prep-XY}^{Y,(2)}$ arise solely due to the presence of fragment Y and X, respectively. This allows Eq. (21) to be simplified as:

$$\Delta E_{int}^{(2)} = \sum_{X>Y} \Delta E_{el-prep}^{XY,(2)} + \sum_{X>Y} \varepsilon^{(X,Y),(2)} = \sum_{X>Y} \Delta E_{int}^{XY,(2)} \tag{26}$$

ΔE_{int}^{coop} can then be obtained in the fp-LED framework by subtracting Eqs. (10) and (26) side by side:

$$\Delta E_{int}^{coop} = \Delta E_{int} - \Delta E_{int}^{(2)} = \sum_{X} \left[\Delta E_{el-prep}^{XY} - \Delta E_{el-prep}^{XY,(2)}\right] + \sum_{X>Y} \left[\varepsilon^{(X,Y)} - \varepsilon^{(X,Y),(2)}\right]$$

$$= \sum_{X} \Delta E_{el-prep}^{XY,coop} + \sum_{X>Y} \varepsilon^{(X,Y),coop} = \sum_{X>Y} \Delta E_{int}^{XY,coop} \tag{27}$$

Eqs. (27) provides a direct means to quantify the overall cooperativity effect and its decomposition into fragment pairwise contributions. Similar to N-body analysis, the two-body and cooperativity terms can be further decomposed into physically meaningful components, including electrostatic, exchange, dispersion, and nondispersion correlation terms.

2.5. Frozen-State LED and Orbital Relaxation

Based on Eqs. (12) and (14), LED partitions the reference-level (typically HF) interaction energy for a fragment pair XY as

$$\Delta E_{int}^{ref} = \Delta E_{el-prep}^{ref} + E_{elstat} + E_{exch} \tag{28}$$

where

$$\Delta E_{el-prep}^{ref} = \Delta E_{el-prep}^{X,ref} + \Delta E_{el-prep}^{Y,ref} \tag{29}$$

Where possible, fragment labels X and Y were omitted in Eqs. (28) and (29) for simplicity. The LED terms in these equations are calculated using fully relaxed localized molecular orbitals of the adduct XY, corresponding to the Ψ_{ref} state. Thus, they inherently incorporate all charge transfer (CT) and polarization effects that arise upon adduct formation, collectively referred to as the "orbital relaxation contribution".

To isolate the orbital relaxation contribution, variational energy decomposition analysis (EDA) schemes define a "frozen state", Ψ_{ref}^{0} (also known as the "promolecular state"), which is devoid of any CT and polarization effects. The difference between the interaction energy of the Ψ_{ref} state, E_{int}^{ref}, and that of the Ψ_{ref}^{0} state, $E_{int}^{ref,0}$, then quantifies the orbital relaxation contribution, $E_{int-o.r.}^{ref}$, *i.e.*,

$$E_{int}^{ref} = E_{int}^{ref,0} + E_{int-o.r.}^{ref} \tag{30}$$

This strategy can also be applied within the LED framework.[29]

In ORCA, Ψ_{ref}^{0} is constructed as the antisymmetrized product of the wave functions of the isolated fragments ($\Psi_{ref,X}$ and $\Psi_{ref,Y}$) at their in-adduct geometries:

$$\Psi_{ref}^{0} = \hat{A}\left(\Psi_{ref,X} \cdot \Psi_{ref,Y}\right) \tag{31}$$

When LED is performed on this frozen state, it partitions the reference-level interaction energy in the frozen state, $E_{int}^{ref,0}$, in a manner analogous to E_{int}^{ref} as given in Eq. (28):

$$E_{int}^{ref,0} = E_{el-prep}^{ref,0} + E_{elstat}^{0} + E_{exch}^{0} \tag{32}$$

$E_{el-prep}^{ref,0}$ represents the energy increase due to the distortion of the electronic structures of the isolated fragments upon antisymmetrization. The sum of $E_{el-prep}^{ref,0}$ and attractive exchange term E_{exch}^{0} in the frozen state corresponds to the "Pauli repulsion" or "exchange-repulsion" term in the variational EDA schemes. E_{elstat}^{0} represents the electrostatic interaction between the electron densities of the fragments X and Y in the frozen state. Thus, it arises from the permanent electrostatic interaction between charges and/or permanent multipoles of the fragments.

Inserting Eqs. (28) and (32) into Eq. (30) and grouping the terms based on their identity, one arrives at the following expressions:

$$E_{el-prep}^{ref} = E_{el-prep}^{ref,0} + E_{el-prep-o.r.}^{ref} \tag{33}$$

$$E_{elstat} = E_{elstat}^{0} + E_{elstat-o.r.} \tag{34}$$

$$E_{exch} = E_{exch}^{0} + E_{exch-o.r.} \tag{35}$$

These equations allow for the analysis of LED interaction energy terms in terms of their frozen and orbital relaxation components, which are sometimes referred to as permanent and inductive components, respectively.

It is worth noting that the difference electron density of Ψ_{ref} and Ψ_{ref}^{0} states serves as a complementary tool to LED for analyzing the origin of orbital relaxation. The "Natural Orbitals for Chemical Valence and Extended Transition State (NOCV-ETS)"[43] method, also available in ORCA,[29] further decomposes this density into binding modes and quantifies their contributions to orbital relaxation. However, this scheme falls outside the current scope.

2.6. CBS Extrapolation

As system size increases, the computational cost of correlation methods grows rapidly, restricting the size of the basis set that can be feasibly applied. To overcome this limitation, several extrapolation schemes have been proposed to estimate energies at the "Complete Basis Set (CBS)" limit using results from finite basis set calculations. Although different choices are possible, the functional dependence of energies on the n-ζ cardinality of the basis set is often considered as:

$$E^{ref,n} = E^{ref} + A \cdot e^{-\alpha\sqrt{n}} \tag{36}$$

for the HF reference part[16] and

$$E^{C,n} = E^{C} + A \cdot n^{-\beta} \tag{37}$$

for the correlation part.[17] Here, $E^{ref,n}$ and $E^{C,n}$ are the reference and correlation energies computed with an n-ζ basis set, while E^{ref} and E^{C} denote their corresponding CBS-limit values.

To determine E^{ref} and E^{C}, Eqs. (36) and (37) can be solved using two basis sets with successive cardinalities n and $m = n + 1$, yielding:

$$E^{ref} = E^{ref,n} + F_{ref} \cdot \left(E^{ref,m} - E^{ref,n}\right) \quad ; \quad F_{ref} = \frac{e^{\alpha\sqrt{m}}}{e^{\alpha\sqrt{m}} - e^{\alpha\sqrt{n}}} \tag{38}$$

$$E^{C} = E^{C,n} + F_{C} \cdot \left(E^{C,m} - E^{C,n}\right) \quad ; \quad F_{C} = \frac{m^{\beta}}{m^{\beta} - m^{\beta}} \tag{39}$$

where the extrapolation coefficients F_{ref} and F_C incorporate the α- and β- dependent terms to simplify the final expressions. These coefficients depend on the choice of basis set family.

Using optimal parameters[44] for α and β, F_{ref} and F_C have been computed for the Dunning (aug-)cc-pVnZ[45–49] and Ahlrichs def2-nVP[50] basis sets, as listed in Table 1.1. It is important to note that, when best-fitting α and β values for the functions in Eqs. (36) and (37) are used, CBS(2/3) extrapolation often leads to poor results. Therefore, a modified set of parameters is typically suggested to improve its predictions (see Table 1.1).

Table 1.1. Optimal α and β parameters and the corresponding F_{ref} and F_C values for CBS(n/m) extrapolations ($m = n + 1$) for the (aug-)cc-pVnZ [n = 2 (D), 3 (T), 4 (Q), 5, …][45–49] and def2-nVP [n = 2 (S), 3 (TZ), 4 (QZ)][50] basis sets.

	CBS(n/m)	α	F_{ref}	β	F_C
(aug-)cc-pVnZ	$n = 2$	4.42	1.325	2.46	1.584
	$n = 3$	5.46	1.301	3.05	1.712
	$n = 4$	5.46	1.308	3.05	2.026
def2-nVP	$n = 2$	10.39	1.038	2.40	1.607
	$n = 3$	7.88	1.378	2.97	1.741

In LEDAW, each decomposed reference and correlation LED term is extrapolated to the CBS limit using Eqs. (38) and (39), respectively. In LEDAW-GUI, selecting CBS(2/3) or CBS(3/4) extrapolation automatically assigns the optimal F_{ref} and F_C parameters for (aug-)cc-pVnZ basis sets (see Table 1.1). To enable extrapolation with other basis set combinations, LEDAW-GUI also allows users to manually specify F_{ref} and F_C parameters.

2.7. CPS Extrapolation

With "TightPNO" settings, DLPNO-CCSD(T) already achieves high accuracy (0.5 kcal/mol) across the entire GMTKN55 benchmark set.[13] However, for weak intermolecular interactions and electronically complex systems —such as those involving transition metals— even tighter T_{CutPNO} values than the TightPNO default (T_{CutPNO} = 10^{-7}) may be required, significantly increasing the computational cost. To overcome this limitation, the "Complete PNO Space (CPS)" extrapolation scheme has been proposed to obtain results at the complete PNO space limit.[14] CPS(X/Y) extrapolation requires two identical TightPNO calculations that differ only in the T_{CutPNO} parameter. The correlation energies obtained from these two computational settings are then extrapolated to the CPS limit for a given basis set as follows:

$$E = E^X + 1.5 \cdot (E^Y - E^X) \quad (40)$$

Here, E^X and E^Y are correlation energies obtained using T_{CutPNO} = 10^{-X} and T_{CutPNO} = 10^{-Y}, respectively, where $Y = X + 1$. E denotes the estimated correlation energy at the CPS limit.

The coefficient 1.5 was determined[14] using the most challenging subsets of GMTKN55[51] and has been found to be valid across all basis set types and sizes, as well as for any X / X+1 value pairs. Thus, using a different coefficient derived from a limited dataset should be avoided, as it may lead to overfitting issues. For this reason, LEDAW-GUI does not allow users to modify the extrapolation coefficient. LEDAW applies Eq. (40) to each LED correlation energy component to determine its value at the CPS limit.

CPS(X/Y) extrapolation is twice as efficient as performing single calculation with T_{CutPNO} = $10^{-(Y+1)}$ while maintaining comparable accuracy.[14] In particular, for a given basis set, CPS(6/7) achieves sub-kJ/mol accuracy (also referred to as benchmark accuracy) by halving the TightPNO error on the most challenging GMTKN55 subsets.[14] Moreover, CPS(6/7) extrapolation eliminates the size dependency of the DLPNO error, enabling accurate results even for large molecular systems.[15]

It is important to emphasize that CBS and CPS extrapolations must be performed simultaneously to account for both basis set incompleteness and PNO incompleteness errors. We have provided compound scripts in the ORCA Github[52] that directly perform the necessary calculations for CBS and/or CPS extrapolations and print the extrapolated reference, correlation, and total energies in a combined ORCA output file from these calculations.[52] However, these scripts were not designed for LED components. Instead, LEDAW enables extrapolations for LED components using a sequence of ORCA output files.

3

GETTING STARTED WITH LEDAW

LEDAW is a Python-based software repository designed to automate and streamline the LED analysis of interaction energies at local coupled-cluster levels within the DLPNO framework. It is organized into two modular components: `ledip` (for pre-processing) and `ledaw` (for post-processing), as outlined in the LEDAW workflow (Figure 3.1).

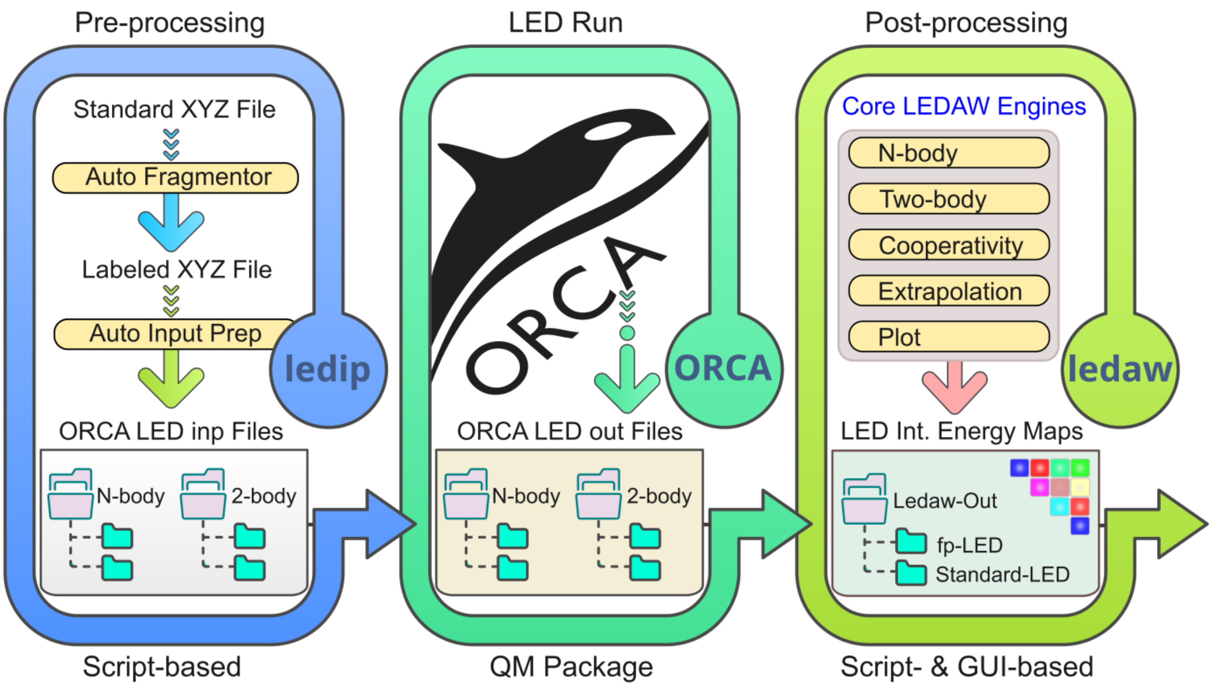


Figure 3.1. Schematic overview of the LED analysis pipeline. `ledip` automates fragmentation and input preparation; ORCA performs the quantum chemical calculations; and `ledaw` post-processes the results into interpretable N-body, two-body, and cooperativity interaction energy matrices and maps, including energy extrapolations to the CPS and CBS limits in the framework of both standard and fp-LED schemes.

The `ledip` package streamlines the pre-processing by providing automatic fragment labeling and input preparation capabilities for N-body and two-body LED calculations. The `ledaw` package facilitates the post-processing of ORCA LED output files using both standard LED and fp-LED schemes, with optional CBS and CPS extrapolations. It also enables the investigation of many-body effects in complex molecular systems. The program generates detailed interaction energy matrices and heat maps, offering in-depth insights into how specific fragment pairs contribute to the overall stability of molecular systems.

This chapter provides an overview of LEDAW's core functionalities, step-by-step instructions for downloading and running the software—either via the GUI or through script-based workflows.

3.1. Getting the Package

To get started, clone or download the full package from the official GitHub repository:

https://github.com/ahmetaltunfatih/LEDAW

If you choose to download the ZIP archive, unzip it and move the extracted contents to your preferred location. The repository includes:

- `ledip_package/` – Core pre-processing `ledip` directory containing source code
- `ledaw_package/` – Core post-processing `ledaw` directory containing source code
- `main.py` – Script for launching LEDAW-GUI directly via Python
- `ledaw.spec` – File for building a standalone LEDAW executable using Pyinstaller
- `docs/` – Documentation and user manual
- `examples/` – Example files for testing and learning, including:
    - `ledip-inputs/` – Python scripts that process XYZ files for automatic fragmentation and LED input file preparations.
    - `orca-outputs/` – Sample ORCA output files for various interaction scenarios.
    - `ledaw-inputs/` – Python scripts that process sample ORCA outputs using LEDAW

3.2. Installation and Running

Before proceeding with the following subsections, ensure that Python is pre-installed in your environment. It is strongly recommended to update your Python installation to the latest version.

3.2.1. Running LEDIP Package (Script-Based Workflow)

The top-level `ledip_package` directory contains two primary Python modules:

- `fragmentation_engine.py` – Detects and labels molecular fragments from the XYZ file of the supersystem.
- `led_input_prep_engine.py` – Generates all required ORCA LED input files for N-body and two-body analyses, both with and without BSSE correction.

Sample Python scripts demonstrating how to run these engines are provided in the "`examples/ledip_inputs/`" directory:

- `run_fragmentor.py`
- `run_inp_prep.py`

See Section 4.5 for the options of these modules and how to run them.

3.2.2. Running LEDAW Package without GUI (Script-Based Workflow)

The `ledaw` package supports a script-based workflow, ideal for users who prefer working with code rather than graphical interface. Several example Python scripts are available in the "`examples/ledaw-inputs/`" directory. These are configured to process the ORCA output files located in "`examples/orca-outputs/`". Running any of these scripts will automatically generate the "`examples/LEDAW-OUT`" directory, along with a subdirectory corresponding to the executed LEDAW job. This subdirectory stores all generated analysis files, such as interaction energy matrices and heat maps.

The script-based workflows are managed by five core engine functions:

- `engine_LED_N_body()`
- `engine_LED_two_body()`
- `cooperativity_engine()`
- `extrapolate_engine()`
- `heatmap_plot_engine_no_gui()`

An additional parser engine, `nbody_to_twobody_engine.py`, automatically aligns fragment labels in the two-body analysis to match those used in the N-body analysis, if both analyses are performed.

To adapt the example scripts for your own systems, simply

- update the paths and filenames of your ORCA output files;
- modify the output path where LEDAW should save results;
- comment out or remove any analysis steps (such as N-body LED, two-body LED, cooperativity, CPS/CBS extrapolations) that are not needed or for which the required ORCA output files are unavailable.

Each example script is well-annotated to support easy understanding and customization. Hence, this document does not include any further discussion of the script-based workflow.

3.2.3. Running LEDAW-GUI Directly with Python

To run LEDAW-GUI directly with Python, navigate to the `LEDAW` directory and execute the following command:

```
python main.py
```

3.2.4. Installing LEDAW-GUI (Executable Generation)

LEDAW-GUI is pre-configured to generate an executable using PyInstaller. If PyInstaller is not already installed on your system, run:

```
pip install pyinstaller
```

Then, navigate to the `LEDAW` directory and execute the following command:

```
pyinstaller ledaw.spec
```

This will generate a standalone LEDAW executable for processing ORCA output files that does not require a Python installation or the source files.

4

LED INPUT PREPARATION AND LEDIP

Since `ledaw` operates on LED output files generated by ORCA, it is essential that the underlying LED calculations are properly set up and that all necessary output files are produced. This section provides sample ORCA input files for LED analysis across various types of interaction problems and highlights key considerations for setting up reliable LED calculations. Preparing such input files can be laborious, especially for multi-fragment systems. To streamline this process, we also provide `ledip` (LED Input Preparator), a companion package that automates input preparation based on a few user-defined settings. Its usage is demonstrated at the end of this chapter.

4.1. LED of Cluster Formation Energy

In this subsection, we present sample ORCA input files for LED of the interaction energy of the boat conformer of the water hexamer, defined as the energy required to bring six water molecules from infinite separation into the given molecular conformation. These inputs are designed to yield results at the DLPNO-CCSD(T_1)/TightPNO/CPS(6/7)/CBS(3/4) level (referred to as the gold DLPNO-CCSD(T) setting[8]) with guidance on how to adapt them for HFLD calculations. It is worth noting that HFLD converges rapidly with respect to both basis set size and PNO settings; therefore, performing computations at the HFLD/TightPNO/CPS(6/7)/CBS(3/4) level is typically unnecessary.

4.1.1. N-Body LED of Water Hexamer Interaction Energy at the Gold DLPNO-CCSD(T_1) Level

Computing the interaction energy of the water hexamer with a single computational setting requires seven ORCA runs in total: one for the supersystem and one for each of the six isolated subsystems (corresponding to individual water molecules). The CPS(6/7) extrapolation involves two sets of computations—one with a looser PNO threshold (T_{CutPNO} = 10^{-6}) and the other with a tighter threshold (T_{CutPNO} = 10^{-7}). Similarly, the CBS(3/4) extrapolation requires computations with two basis sets: a smaller basis set (aug-cc-pVTZ) and a larger basis set (aug-cc-pVQZ). Therefore, to perform an N-body LED analysis of the water hexamer at the DLPNO-CCSD(T_1)/TightPNO/CPS(6/7)/CBS(3/4) level, a total of 7 × 2 × 2 = 28 electronic structure runs with ORCA are needed, as illustrated in Figure 4.1.

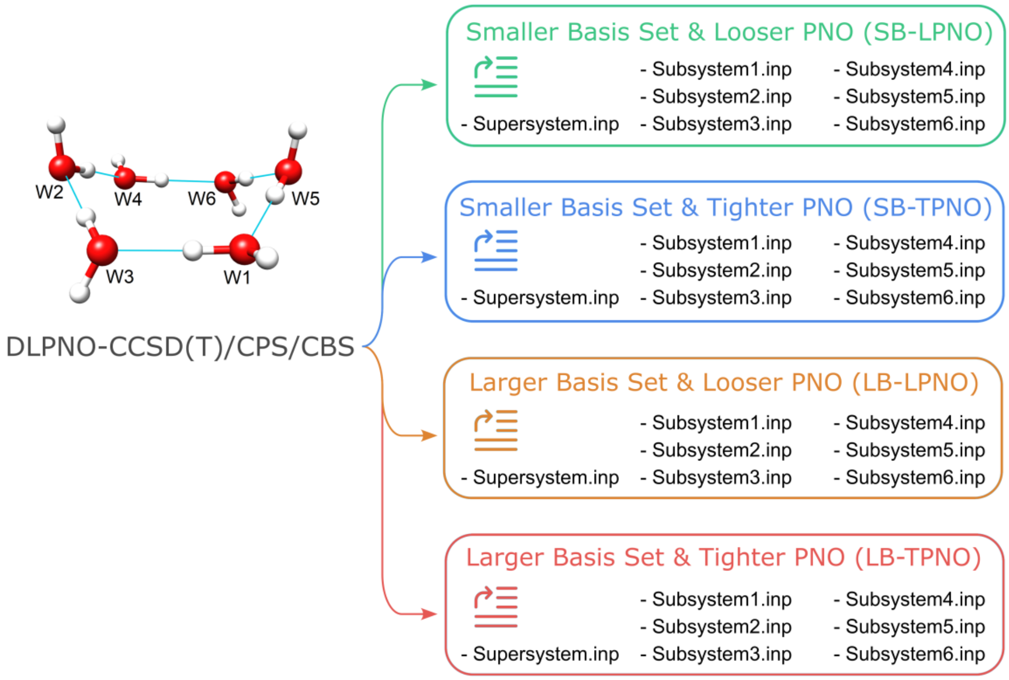


Figure 4.1. Four computational settings required for simultaneously performing CPS and CBS extrapolations of the computed water hexamer interaction energies for N-bod LED analysis. Each setting necessitates seven DLPNO-CCSD(T) runs with ORCA. The corresponding ORCA input files for these runs can be automatically generated using the `ledip` package based on the XYZ file of the supersystem (see Section 4.5). From the outputs of these ORCA runs, LEDAW further generates intermediate LED results for CPS extrapolation with the smaller basis set (SB-CPS) and the larger basis set (LB-CPS), CBS extrapolation with the looser PNO (CBS-LPNO) and the tighter PNO (CBS-TPNO), as well as the final CPS-CBS extrapolated interaction energy. Fragment labels were intentionally randomized to demonstrate LEDAW's relabeling feature (see the next chapter).

4.1.1.1. Supersystem Inputs

As shown in Figure 4.1, LED computation on the supersystem is required for each of the four settings. For the **SB-LPNO** setting (aug-cc-pVTZ; $T_{CutPNO} = 10^{-6}$), a sample input is as follows:

```
! DLPNO-CCSD(T1) aug-cc-pVTZ RIJCOSX DefGrid3 def2/J aug-cc-pVTZ/C
! VeryTightSCF TightPNO LED

%mdci TCutPNO 1e-6 end

*xyz 0 1
O(1)    -1.753042425     0.416462959     0.356105280
H(1)    -1.120933301    -0.183098298     0.812596940
H(1)    -2.593310567    -0.048459739     0.359102074
O(2)     2.482542995    -1.248988714     0.449488905
H(2)     2.616828786    -1.798226083    -0.326968465
H(2)     2.828107886    -0.362220961     0.204648083
O(3)     0.012631031    -1.241255898     1.573836894
H(3)     0.908618001    -1.252949881     1.168524322
H(3)     0.168950409    -1.135399007     2.515952529
O(4)     3.364853284     1.252522284    -0.119963272
```

```
H(4)      2.731853532      1.891821037     -0.517486897
H(4)      4.212012919      1.443946099     -0.529713339
O(5)     -0.845377446      1.902266646     -1.724481185
H(5)     -1.198971324      1.335278306     -1.003913817
H(5)     -0.953209570      1.388839038     -2.529056860
O(6)      1.597560181      2.978636709     -1.236245995
H(6)      1.417961949      3.808511891     -0.786488687
H(6)      0.711446199      2.593888593     -1.420550970
*
```

To generate inputs for the other three settings, modify this sample input as follows:

- **SB-TPNO:** Replace "`TCutPNO 1e-6`" in the `mdci` block with "`TCutPNO 1e-7`", or simply remove the `mdci` block as `1e-7` is the default for TightPNO.
- **LB-LPNO:** Replace "`aug-cc-pVTZ`" with "`aug-cc-pVQZ`".
- **LB-TPNO:** Replace "`TCutPNO 1e-6`" and "`aug-cc-pVTZ`" with "`TCutPNO 1e-7`" and "`aug-cc-pVQZ`".

In this example, the `RIJCOSX` approximation is used in the reference part. This choice is particularly advantageous for large systems. While the `RIJK` approximation is slightly more accurate, it can become prohibitively expensive for LED calculations on large molecular clusters.

In this input, fragment labels (shown also on Figure 4.1) are specified in parentheses following each atomic symbol in the `xyz` block. Since all fragments in this example are standard fragments, *i.e.*, interacting with each other noncovalently, the user actually does not need to assign fragment labels unless there is a specific reason to do so. If no labels are provided, the `LED` keyword triggers automatic fragment labeling based on atomic connectivity. For alternative fragment labeling methods, such as providing index list of atom at the end of the input, or through external coordinate files (supported from ORCA 6.1+), or specifying labels for only certain fragments, please refer to the corresponding ORCA manuals.

If labels are specified manually, it is strongly recommended to begin numbering from 1 and assign them consecutively without gaps, in accordance with the automatic labeling scheme. In fact, any label may be assigned any fragment, but the ordered label list should proceed without gaps.

For instance, using labels 5 and 7 for a two-fragment system prompts ORCA to retain placeholders for fragments 1–4 and 6 in the output, unnecessarily complicating the output structure. Moreover, in some ORCA versions, LED calculations may crash during the decomposition of strong-pairs energy when non-consecutive labels are used.

4.1.1.2. Subsystem Inputs for BSSE-Uncorrected Interaction Energy

To compute BSSE-uncorrected interaction energy and its LED components, calculations on all isolated subsystems are also required. These calculations must use the exact same computational settings as the supersystem. However, the `LED` keyword must be removed from the subsystem inputs because LED calculations require at least two real fragments, whereas subsystems here correspond to individual water molecules. Below is a sample input for **fragment 1** of the water hexamer using the **SB-LPNO** setting:

```
! DLPNO-CCSD(T1) aug-cc-pVTZ RIJCOSX DefGrid3 def2/J aug-cc-pVTZ/C
! VeryTightSCF TightPNO

%mdci TCutPNO 1e-6 end

*xyz 0 1
O    -1.753042425     0.416462959     0.356105280
H    -1.120933301    -0.183098298     0.812596940
H    -2.593310567    -0.048459739     0.359102074
*
```

For the inputs of the remaining five fragments using the **SB-LPNO** setting, just replace the coordinates of fragment 1 with those of the respective fragment in the sample input above.

To prepare subsystem inputs for the other three settings necessary for simultaneous CPS and CBS extrapolations (i.e., SB-TPNO, LB-LPNO, and LB-TPNO), apply the same modifications as described in the supersystem section.

LEDAW uses the fragment labels in the supersystem ORCA output file (but not those in the subsystem outputs). It aligns subsystem fragments with supersystem fragments by comparing their coordinates in ORCA outputs. Therefore, LEDAW will not function properly if the coordinates of a given fragment in the supersystem and subsystem files differ beyond a certain tolerance, even if the fragments are structurally identical but spatially shifted. Ensuring consistent atomic coordinates between the supersystem and subsystems is therefore essential for a successful LEDAW run.

4.1.1.3. Subsystem Inputs for BSSE-Corrected Interaction Energy

To compute BSSE-corrected interaction energy and its LED components, calculations must be performed on all isolated subsystems using exactly the same basis set as that used for the supersystem. Hence, each subsystem input must include the full set of atomic coordinates from the supersystem in the `xyz` block. The subsystem under consideration must be treated as *real*, while all other fragments must be designated as *ghost* by placing colon sign ":" after each of the respective atomic symbol in the `xyz` block.

Below is a sample input for **fragment 3** of the water hexamer using the **SB-LPNO** setting. In this example, fragment labels are included for clarity but are not strictly necessary, as previously explained.

```
! DLPNO-CCSD(T1) aug-cc-pVTZ RIJCOSX DefGrid3 def2/J aug-cc-pVTZ/C
! VeryTightSCF TightPNO

%mdci TCutPNO 1e-6 end

*xyz 0 1
O:(1)    -1.753042425     0.416462959     0.356105280
H:(1)    -1.120933301    -0.183098298     0.812596940
H:(1)    -2.593310567    -0.048459739     0.359102074
O:(2)     2.482542995    -1.248988714     0.449488905
H:(2)     2.616828786    -1.798226083    -0.326968465
H:(2)     2.828107886    -0.362220961     0.204648083
O(3)      0.012631031    -1.241255898     1.573836894
H(3)      0.908618001    -1.252949881     1.168524322
H(3)      0.168950409    -1.135399007     2.515952529
O:(4)     3.364853284     1.252522284    -0.119963272
H:(4)     2.731853532     1.891821037    -0.517486897
```

```
H:(4)     4.212012919     1.443946099    -0.529713339
O:(5)    -0.845377446     1.902266646    -1.724481185
H:(5)    -1.198971324     1.335278306    -1.003913817
H:(5)    -0.953209570     1.388839038    -2.529056860
O:(6)     1.597560181     2.978636709    -1.236245995
H:(6)     1.417961949     3.808511891    -0.786488687
H:(6)     0.711446199     2.593888593    -1.420550970
*
```

Note that the ":" sign is omitted for fragment 3 only, as this input is intended for fragment 3. To generate inputs for other fragments, simply add ":" for fragment 3 and remove it from the respective target fragment. All other computational settings for the simultaneous CPS and CBS extrapolations necessitate following the modifications as already described for the supersystem input.

4.1.1.4. ORCA Output Files for the N-Body LED

To perform an N-body LED analysis of the water hexamer interaction energy at the DLPNO-CCSD(T_1)/TightPNO/CPS(6/7)/CBS(3/4) level, ORCA must be run for all inputs described in the preceding subsections—28 for BSSE-uncorrected interaction energy and 28 for BSSE-corrected interaction energy, with 4 supersystem runs common to both. Therefore, ensure that all required calculations are completed successfully and that the corresponding output files are available.

LEDAW is distributed with the ORCA output files necessary to compute the BSSE-uncorrected DLPNO-CCSD(T_1)/TightPNO/CPS(6/7)/CBS(3/4) interaction energy of the boat conformer of the water hexamer. These files allow users to easily setup similar LED calculations and practice using LEDAW. Note that the computational settings in these sample output files differ slightly from those described above, using auto-generated auxiliary basis sets instead of the standard "/J" and "/C" basis sets.

4.1.2. Two-Body LED of Water Hexamer Interaction Energy at the Gold DLPNO-CCSD(T_1) Level

To evaluate the influence of the chemical environment on fragment pairs within a supersystem, additional relatively inexpensive computations must be performed—specifically, calculations on all isolated fragment pairs across the subsystems together with their corresponding individual fragment calculations. Therefore, conducting a two-body LED analysis for the interaction energy of the water hexamer requires C(6, 2) = 6 × 5 / 2 = 15 dimer (two-body) calculations using a single computational setting, in addition to isolated fragment (one-body) calculations.

If BSSE-uncorrected interaction energy is considered, then only a single one-body calculation per fragment is required, amounting to six in total for the water hexamer. Each resulting one-body output file is reused for calculating the interaction energy of all dimers involving that fragment. These one-body outputs are identical for the water hexamer case to those used in the BSSE-uncorrected N-body LED analysis, so no additional one-body jobs are needed.

In contrast, if BSSE-corrected interaction energy is considered, then for each dimer, two fragment calculations are required—each having one real and one ghost fragments, amounting to 15 × 2 = 30 one-body calculations. Hence, for a single computational setting, the two-body LED analysis of the water hexamer requires:

- 21 ORCA runs if BSSE is not corrected (15 dimers + 6 monomers)
- 45 ORCA runs if BSSE is corrected (15 dimers + 30 monomers)

Performing DLPNO-CCSD(T_1)/TightPNO/CPS(6/7)/CBS(3/4) level computations, which involve four distinct computational settings, thus requires:

- 84 runs for BSSE-uncorrected analysis (21 × 4)
- 180 runs for BSSE-corrected analysis (45 × 4)

The `ledip` package automates the otherwise laborious task of generating all corresponding ORCA input files directly from the XYZ file of the supersystem (see Section 4.5). Although the number of calculations may appear substantial, the overall computational cost remains modest, as each job involves no more than two fragments.

We have described in the previous section how to modify inputs for different computational settings and provided BSSE-uncorrected fragment inputs. Therefore, in the following, we consider only the SB-LPNO setting with BSSE correction. Below are sample inputs for the BSSE-corrected interaction energy calculation of **fragment pair 1–2** in the water hexamer:

Two-body input for pair 1-2

```
! DLPNO-CCSD(T1) aug-cc-pVTZ RIJCOSX DefGrid3 def2/J aug-cc-pVTZ/C
! VeryTightSCF TightPNO LED

%mdci TCutPNO 1e-6 end

*xyz 0 1
O(1)     -1.753042425     0.416462959     0.356105280
H(1)     -1.120933301    -0.183098298     0.812596940
H(1)     -2.593310567    -0.048459739     0.359102074
O(2)      2.482542995    -1.248988714     0.449488905
H(2)      2.616828786    -1.798226083    -0.326968465
H(2)      2.828107886    -0.362220961     0.204648083
*
```

One-body input for fragment 1 of pair 1-2 (with fragment 2 as ghost)

```
! DLPNO-CCSD(T1) aug-cc-pVTZ RIJCOSX DefGrid3 def2/J aug-cc-pVTZ/C
! VeryTightSCF TightPNO

%mdci TCutPNO 1e-6 end

*xyz 0 1
O(1)     -1.753042425     0.416462959     0.356105280
H(1)     -1.120933301    -0.183098298     0.812596940
H(1)     -2.593310567    -0.048459739     0.359102074
O:(2)     2.482542995    -1.248988714     0.449488905
H:(2)     2.616828786    -1.798226083    -0.326968465
H:(2)     2.828107886    -0.362220961     0.204648083
*
```

One-body input for fragment 2 of pair 1-2 (with fragment 1 as ghost)

```
! DLPNO-CCSD(T1) aug-cc-pVTZ RIJCOSX DefGrid3 def2/J aug-cc-pVTZ/C
! VeryTightSCF TightPNO

%mdci TCutPNO 1e-6 end
```

```
*xyz 0 1
O:(1)    -1.753042425     0.416462959     0.356105280
H:(1)    -1.120933301    -0.183098298     0.812596940
H:(1)    -2.593310567    -0.048459739     0.359102074
O(2)      2.482542995    -1.248988714     0.449488905
H(2)      2.616828786    -1.798226083    -0.326968465
H(2)      2.828107886    -0.362220961     0.204648083
*
```

Analogous inputs must also be prepared for the remaining 14 fragment pairs by updating the coordinates accordingly. This will result in 15 × 3 = 45 inputs for the SB-LPNO setting alone.

Fragment labels are included in the above inputs solely for clarity; if not provided, ORCA assigns them automatically. However, if labels are manually specified for two-body inputs, it is strongly suggested to label them as 1 and 2 in all inputs, as already explained previously.

Importantly, LEDAW retrieves fragment labels exclusively from the supersystem file, regardless of the labels used in two-body or subsystem inputs. Fragment matching is performed based on coordinates, not labels. Therefore, it is crucial to ensure that fragment coordinates are identical across N-body and two-body LED analyses for a successful joint LEDAW run aimed at evaluating cooperativity effects.

LEDAW is distributed with the ORCA output files required to perform the BSSE-uncorrected two-body LED analysis of the water hexamer at the DLPNO-CCSD(T_1)/TightPNO/CPS(6/7)/CBS(3/4) level. The computational settings in these files are fully aligned with those used in the corresponding N-body calculations.

4.1.3. LED of Water Hexamer Interaction Energy at the HFLD Level

The explanations provided above for the BSSE-uncorrected and BSSE-corrected N-body and two-body LED analyses also apply to HFLD computations. Therefore, this section focuses only on how to modify DLPNO-CCSD(T) inputs for HFLD calculations. A sample HFLD input for the water hexamer supersystem using the SB-LPNO setting is given below:

```
! HFLD aug-cc-pVTZ RIJCOSX DefGrid3 def2/J aug-cc-pVTZ/C
! VeryTightSCF TightPNO

%mdci TCutPNO 1e-6
      DoLedHF True
      end

*xyz 0 1
O(1)    -1.753042425     0.416462959     0.356105280
H(1)    -1.120933301    -0.183098298     0.812596940
H(1)    -2.593310567    -0.048459739     0.359102074
O(2)     2.482542995    -1.248988714     0.449488905
H(2)     2.616828786    -1.798226083    -0.326968465
H(2)     2.828107886    -0.362220961     0.204648083
O(3)     0.012631031    -1.241255898     1.573836894
H(3)     0.908618001    -1.252949881     1.168524322
H(3)     0.168950409    -1.135399007     2.515952529
O(4)     3.364853284     1.252522284    -0.119963272
H(4)     2.731853532     1.891821037    -0.517486897
H(4)     4.212012919     1.443946099    -0.529713339
O(5)    -0.845377446     1.902266646    -1.724481185
H(5)    -1.198971324     1.335278306    -1.003913817
H(5)    -0.953209570     1.388839038    -2.529056860
O(6)     1.597560181     2.978636709    -1.236245995
```

```
H(6)     1.417961949     3.808511891    -0.786488687
H(6)     0.711446199     2.593888593    -1.420550970
*
```

Hence, to adapt a DLPNO-CCSD(T)/LED supersystem input for HFLD, one simply needs to replace the method name with `HFLD` and enable HF/LED decomposition by setting `DoLedHF = True` in the `mdci` block (default is `False`). The `LED` keyword is not required in an HFLD input, as LED is inherently part of the method.

Alternatively, HFLD can be invoked via the multilevel DLPNO-CCSD implementation by including a series of keywords in the `mdci` block, such as `HFFragInter` directives to exclude intra-subsystem correlation. In this case, the final energy printed in the ORCA output includes residual correlation components and does *not* represent the pure HFLD energy. To obtain the correct HFLD energy, the reference and dispersion energies must be summed manually. The equivalent multilevel HFLD input to the above example is shown below:

```
! DLPNO-CCSD aug-cc-pVTZ RIJCOSX DefGrid3 def2/J aug-cc-pVTZ/C
! VeryTightSCF TightPNO LED

%mdci TCutPNO 1e-6
      LocMet FB
      LocTol 1.0e-6
      LocRandom 1
      HFFragInter {1 1} {2 2} {3 3} {4 4} {5 5} {6 6}
      end
```

Regardless of the method name printed in ORCA outputs, when HFLD is specified as the method name in a LEDAW run, only the dispersion component is retained in the correlation part, ensuring consistency with the original HFLD definition.

By default, HFLD energy is obtained by summing the SCF and dispersion energies. However, LED dissects the reference `E(0)` energy rather than the SCF energy. For closed-shell systems, SCF (RHF) and `E(0)` energies are identical. Minor variations may arise if an RI approach is not specified (*e.g.*, via "`! RIJK`" or "`! RIJCOSX`"), but these are typically negligible. Therefore, for the water hexamer, RHF calculations alone are sufficient for subsystems. This does not apply to multifragment subsystems, which will be discussed in the next section. Below is a sample input for **fragment 1** of the water hexamer (without BSSE correction):

```
! HF aug-cc-pVTZ RIJCOSX DefGrid3 def2/J VeryTightSCF

*xyz 0 1
O    -1.753042425     0.416462959     0.356105280
H    -1.120933301    -0.183098298     0.812596940
H    -2.593310567    -0.048459739     0.359102074
*
```

For open-shell systems, the SCF and `E(0)` energies correspond to UHF and QRO, respectively. Consequently, LED dissects the QRO/HFLD energy rather than the UHF/HFLD. Therefore, unlike closed-shell subsystems, HF calculations alone are insufficient for open-shell subsystems, as the QRO energy is required. A practical way to obtain the QRO energy is to run a DLPNO-CCSD calculation with very loose correlation settings. A sample input for such a calculation is as follows:

```
! DLPNO-CCSD aug-cc-pVTZ RIJCOSX DefGrid3 def2/J aug-cc-pVTZ/C
! VeryTightSCF LoosePNO

%mdci maxiter 0 end

*xyz 0 3
...
*
```

Note that the `maxiter 0` directive in the `mdci` block is used to skip unnecessary coupled-cluster iterations. Since the correlation energy is not required for subsystems, this ORCA run may even be terminated after `E(0)` is printed.

It is worth noting that, if both DLPNO-CCSD(T) and HFLD calculations are to be performed, separate subsystem jobs for HFLD are unnecessary, as the DLPNO-CCSD(T) subsystem outputs already include the required SCF or `E(0)` energies.

LEDAW allows the use of HF energy instead of `E(0)` for subsystems when the method is set to HFLD. However, as noted above, this option is only appropriate for closed-shell systems.

All of the above considerations also apply to two-body HFLD/LED analyses, except that, instead of a supersystem input, analogous inputs must be prepared for all fragment pairs.

4.2. LED of Cluster-Cluster Interaction Energy

In this subsection, we provide sample ORCA input files for LED analysis of the interaction energy between the two strands of a three-nucleobase-long DNA segment illustrated in Figure 2.2. The inputs are designed to yield results at both the DLPNO-CCSD(T) and HFLD levels, using the NormalPNO* settings and the def2-TZVP(-f) basis set. The effect of the water environment is incorporated implicitly via the CPCM model. The corresponding ORCA output files are distributed with the LEDAW package.

We also distribute ORCA output files required for lattice energy calculations using HFLD with the same settings as in the DNA example. In this case, the subsystems comprise the central monomer (a single fragment, as discussed in Section 4.1) and the surrounding crystal cluster (a multifragment system, as discussed in this section). Since single and multifragment subsystem cases are covered in Section 4.1. as well as in the present section, no further explanation is warranted for this crystal example.

Two-body LED analysis will not be discussed further in this section, as the general procedures and input preparation guidelines have already been outlined in the preceding section. The same considerations apply here, with the exception that only fragment pairs in which the two fragments reside on different DNA strands are considered, rather than all possible fragment pairs.

4.2.1. LED of DNA Duplex Formation Energy at the DLPNO-CCSD(T) Level

Below is a sample ORCA input for performing a DLPNO-CCSD(T)/LED calculation on the DNA supersystem with the settings described above:

```
! DLPNO-CCSD(T) def2-TZVP(-f) RIJCOSX DefGrid3 def2/J def2-TZVP/C
! VeryTightSCF CPCM(water) NormalPNO LED

%mdci TCutPairs 1e-5 end
```

```
*xyz 0 1
 H(1)      10.634700344584      13.691281144512     -11.875553166247
 N(1)       9.980106156592      13.839587738160     -11.071433575066
 C(1)      10.442864004832      13.875896090101      -9.798268811879
 H(1)      11.504832150180      13.715386729876      -9.677066287960
 C(1)       9.606316071622      14.082566153675      -8.757444583846
 H(1)       9.969796490105      14.122858111485      -7.746934660848
 C(1)       8.213760646535      14.241454159912      -9.044751776248
 N(1)       7.327081712146      14.401901044136      -8.074002131001
 H(1)       7.652431838508      14.636728247003      -7.150492447526
 H(1)       6.356598650077      14.618122743997      -8.319845047378
 N(1)       7.770953959427      14.126729133636     -10.296510374266
 C(1)       8.599854136225      13.955873478910     -11.327889587706
 O(1)       8.191469311313      13.869760857544     -12.487179766903
 H(2)       7.908971458264      16.330584347190     -14.914226197121
 N(2)       7.881940637725      16.532665928810     -13.880511672013
 C(2)       9.012048768510      16.770206523525     -13.152689461214
 H(2)       9.940538578336      16.685266706211     -13.698351026605
 C(2)       9.004919716563      17.074088156969     -11.839690594854
 C(2)      10.229508549268      17.422595087270     -11.055129656290
 H(2)      11.104663398618      17.455282888334     -11.696506986450
 H(2)      10.077971944522      18.396373650562     -10.593668745583
 H(2)      10.390332193916      16.701336107263     -10.255896978574
 C(2)       7.728677472520      17.111302212168     -11.152199754353
 O(2)       7.591576070140      17.345499551464      -9.956268414904
 N(2)       6.645820288608      16.805693851400     -11.932197008459
 H(2)       5.695108359505      16.933012057655     -11.497671886556
 C(2)       6.643971848457      16.537517940386     -13.265921081701
 O(2)       5.608659448998      16.284373997746     -13.862115698184
 H(3)       4.167079527237      19.199246063134     -16.022543736994
 N(3)       4.720709708947      19.310013304079     -15.150325886002
 C(3)       6.064240770076      19.215142752868     -14.931704323361
 H(3)       6.747843821168      18.967933282822     -15.718663332453
 N(3)       6.412548423676      19.456483831030     -13.703531446170
 C(3)       5.232781972225      19.735449439261     -13.049658288826
 C(3)       4.949492015809      20.145779207010     -11.720076857315
 O(3)       5.745437995832      20.316096696627     -10.796304698950
 N(3)       3.593297648465      20.377473357167     -11.520990626887
 H(3)       3.323052692831      20.746411963262     -10.587381622814
 C(3)       2.622889432685      20.208130250514     -12.467641661931
 N(3)       1.351243114016      20.384866184300     -12.095051028829
 H(3)       1.138815575017      20.797025787658     -11.187898890469
 H(3)       0.665646529482      20.443543657416     -12.828192563467
 N(3)       2.872666242150      19.837350187609     -13.703978842593
 C(3)       4.173814909084      19.635730662889     -13.950649270381
 H(4)       0.607492786974      13.693167256443     -12.498905296936
 N(4)       1.156233913621      13.976335931793     -11.654346631242
 C(4)       0.719515992071      14.493104455789     -10.463418891368
 H(4)      -0.320190226981      14.652465882135     -10.253396768309
 N(4)       1.675522233408      14.756439112108      -9.624659463980
 C(4)       2.833919441760      14.430973751698     -10.293445908760
 C(4)       4.205441626210      14.518429914241      -9.917743216750
 O(4)       4.678969477231      14.902137378802      -8.847036060363
 N(4)       5.046351902043      14.044212601799     -10.915024253057
 H(4)       6.072714102757      14.111198911849     -10.723202969726
 C(4)       4.637211663222      13.564345980913     -12.125004367406
 N(4)       5.571436643365      13.009014240888     -12.919396141151
 H(4)       6.536208806688      13.328415372914     -12.798515557053
 H(4)       5.278456082314      12.891881947995     -13.876688905133
 N(4)       3.372577813131      13.480706123713     -12.484778526663
 C(4)       2.516506103177      13.957107881401     -11.567141462038
 H(5)      -0.427318750124      17.859731737670     -10.029201555848
 N(5)       0.505872319628      18.084321359603      -9.624732567559
 C(5)       0.836583758659      18.604519628671      -8.407322768699
```

```
 H(5)       0.092411961428      18.948262192878      -7.715100794658
 N(5)       2.115426925402      18.643562050428      -8.177257646546
 C(5)       2.692395320505      18.137378704062      -9.318466049416
 C(5)       4.023866632235      17.839636438080      -9.678850958220
 N(5)       5.069903086692      17.996703370707      -8.878260648241
 H(5)       4.919055989330      18.448434690564      -7.993811348555
 H(5)       6.012493323398      17.797866055580      -9.225623461520
 N(5)       4.208096816167      17.308977601422     -10.903952927055
 C(5)       3.171511385838      17.075799836042     -11.696070915968
 H(5)       3.416948001374      16.659607513112     -12.669038097596
 N(5)       1.893387444935      17.271543482840     -11.450623229708
 C(5)       1.692338416848      17.804060912579     -10.238211580065
 H(6)       0.347959720865      22.070953898527      -7.231147376111
 N(6)       1.367130517057      22.050245111897      -7.460759445434
 C(6)       2.293054934819      22.474694200979      -6.575605167157
 H(6)       1.905330062598      22.892493241868      -5.657222755113
 C(6)       3.613987194667      22.359676299416      -6.836971827951
 H(6)       4.357477801351      22.686127035235      -6.131652395756
 C(6)       3.986574767236      21.774827021668      -8.089632503708
 N(6)       5.248199676647      21.489136249217      -8.363368075578
 H(6)       5.974584821090      21.754378375091      -7.719014977155
 H(6)       5.487376252505      21.069865859566      -9.271452130257
 N(6)       3.057799105691      21.473978059250      -8.997786542956
 C(6)       1.758387335929      21.635696482051      -8.749410306360
 O(6)       0.886565761830      21.396547045533      -9.589416228698
*
```

The `NormalPNO*` designation refers to the use of `NormalPNO` settings in combination with the `TCutPairs` threshold adopted from the `TightPNO` settings.[7] Therefore, in addition to requesting the `NormalPNO` settings, the "`TCutPairs 1e-5`" directive must be included within the `mdci` block. An explicit request for `NormalPNO` is not needed, as it is the default for DLPNO-CCSD(T). It is included in the above example solely for clarity.

To compute the inter-strand interaction energy, subsystem 1 and subsystem 2 must be defined as strand K (fragments 1–3) and strand L (fragments 4–6), as shown in Figure 2.2. If BSSE correction is requested, the subsystem inputs can be prepared by modifying the supersystem input as follows:

- *Subsystem 1 input:* Prefix fragment labels (4), (5), and (6) with a "`:`" sign to designate them as ghost atoms.
- *Subsystem 2 input:* Prefix fragment labels (1), (2), and (3) with a "`:`" sign to designate them as ghost atoms.

If BSSE correction is not required, the ghost fragments must be entirely removed from the respective subsystem input files. As a result, the subsystem 1 input contains only the strand K fragments, while the subsystem 2 input contains only the strand L fragments. For a cleaner ORCA output, the fragments in the subsystem 2 input file should be relabeled as (1), (2), and (3) instead of (4), (5), and (6).

4.2.2. LED of DNA Duplex Formation Energy at the HFLD Level

Below is a sample ORCA input for performing an HFLD/LED calculation on the DNA supersystem with the settings described above:

```
! HFLD def2-TZVP(-f) RIJCOSX DefGrid3 def2/J def2-TZVP/C
! VeryTightSCF CPCM(water)

%mdci DoLedHF True
      HFFragInter {1 2} {1 3} {2 3} {4 5} {4 6} {5 6}
```

```
      end

*xyz 0 1
...
*
```

`NormalPNO*` settings are the default for HFLD and are therefore not specified in this example. However, they could also be explicitly requested, as shown in the previous subsection. For coordinates and fragment labels omitted here for brevity, see the previous subsection as well.

By definition, HFLD neglects intra-subsystem correlation, keeping the subsystem treatment at the reference level. However, fragment assignment in ORCA inputs is based on fragment labels rather than their associated subsystem grouping. As a result, default HFLD implementation in ORCA assumes each fragment as a separate subsystem. To explicitly define which fragment pairs are to be excluded from correlation, the `HFFragInter` directive must be used. In the supersystem input above, this directive sets the correlation energy among strand K fragments (1–3) and among strand L fragments (4–6) to zero. It is not necessary to specify intra-fragment correlations explicitly, as these are already excluded by the HFLD keyword.

In this DNA example, since each subsystem consists of multiple fragments, standalone RHF or QRO calculations for the subsystems are insufficient. RHF/LED or QRO/LED analysis must also be performed to enable pairwise decomposition of the interaction energy. If both DLPNO-CCSD(T) and HFLD calculations are to be performed, the required reference level LED analyses are already included in the DLPNO-CCSD(T) subsystem outputs, and no additional subsystem jobs are needed for HFLD. Otherwise, because LED is implemented exclusively within the DLPNO framework and cannot be invoked at the reference level alone, a low-cost DLPNO-CCSD or HFLD correlation calculation can be submitted to efficiently obtain the reference level LED data for the subsystems, as illustrated below:

```
! HFLD def2-TZVP(-f) RIJCOSX DefGrid3 def2/J def2-TZVP/C
! VeryTightSCF CPCM(water) LoosePNO

%mdci DoLedHF True
      maxiter 0
      end

*xyz 0 1
...
*
```

Here the `xyz` block must include subsystem coordinates, as described in the previous subsection. The `maxiter 0` directive in the `mdci` block is for skipping unnecessary coupled-cluster iterations.

When setting up a low-cost correlation calculation, avoid cutting the correlation space too aggressively: at least one electron pair must remain to perform a successful ORCA run that prints the reference LED data. In this example, LoosePNO is requested. However, if no electron pair remains, slightly tighter PNO settings may be needed.

For large systems, selectively excluding some interfragment correlations might help reduce the computational cost. However, specifying *all* interfragment pairs in the

“HFFragInter” directive will exclude *all* electron pairs, thereby disabling LED analysis entirely.

If the subsystem labels are 1–3, the “HFFragInter {1 1} {2 2} {3 3}” directive is automatically applied when using HFLD keyword. As intra-fragment correlation is the most computational demanding part, this directive reduces the computational cost enormously. Below is an analogous input to the above HFLD subsystem example, using the multi-level DLPNO-CCSD implementation:

```
! DLPNO-CCSD def2-TZVP(-f) RIJCOSX DefGrid3 def2/J def2-TZVP/C
! VeryTightSCF CPCM(water) LoosePNO LED

%mdci HFFragInter {1 1} {2 2} {3 3}
      LocMet FB
      LocTol 1.0e-6
      LocRandom 1
      maxiter 0
      end

*xyz 0 1
...
*
```

Finally, it is worth reemphasizing that when HFLD is specified as the method in a LEDAW run, redundant correlation terms in ORCA outputs are automatically discarded.

4.3. LED with Implicit Solvation and QM/MM Schemes

4.3.1. LED with Implicit Solvation Schemes

Implicit solvation schemes can be requested by adding relevant keywords in the simple input line. For example, to specify water as solvent:

```
! CPCM(water)
```

and

```
! SMD(water)
```

For a complete list of available solvents, options, and the syntax for different ORCA versions, please refer to the corresponding ORCA manual.

When these schemes are applied, the reference energy (RHF or QRO) includes additional contributions: a “dielectric” term of electrostatic origin and, in the case of SMD, a cavity-dispersion-solvent-structure (CDS) term of nonelectrostatic origin. In ORCA outputs, these solvation contributions to the SCF energy are printed just after SCF iterations:

```
               *****************************************************
               *                     SUCCESS                       *
               *           SCF CONVERGED AFTER  12 CYCLES          *
               *****************************************************
…
----------------
TOTAL SCF ENERGY
----------------
…
```

```
Total Energy         :      -2781.16658238174341 Eh          -75679.39017 eV
…
CPCM Dielectric      :         -0.19788878834679 Eh              -5.38483 eV
SMD CDS (Gcds)       :          0.00958762146934 Eh               0.26089 eV
```

Similarly, depending on the chosen coupled-cluster implicit solvation model, a small dielectric contribution may also appear in the correlation energy. In ORCA outputs, this contribution is printed just after coupled-cluster iterations:

```
               --- The Coupled-Cluster iterations have converged ---

----------------------
COUPLED CLUSTER ENERGY
----------------------

E(0)                                       ...  -2781.166582545
E(CORR)(strong-pairs)                      ...     -9.552366935
E(CORR)(weak-pairs)                        ...     -0.041630620
E(CORR)(corrected)                         ...     -9.593997555
C-PCM corr. term (included in E(CORR))     ...     -0.000362075
E(TOT)                                     ...  -2790.760580100
```

LED analysis is performed with the implicit solvation terms removed from the reference and correlation energies, while their polarization effects on the wave function are still included. For the example above, LED section of the output reports the total reference energy as:

```
Total REF. energy                    =     -2780.978281071083
```

Note that when the CPCM dielectric and CDS terms from the reference part are added to this value, the `E(0)` energy is exactly reproduced.

LEDAW handles implicit solvation terms automatically, so no extra care is required from the user. These terms are excluded from the reference and correlation energies and reported as a separate fragment-pairwise interaction component `SOLV` in the LEDAW summary files. The intermediate LEDAW files additionally provide the individual reference dielectric, correlation dielectric, and SMD interaction energy matrices.

4.3.2. LED with QM/MM

In additive QM/MM schemes, the energy of the entire system ($E^{QM/MM}$) is expressed as

$$E^{QM/MM} = E^{QM} + E^{MM} \tag{41}$$

Here E^{MM} is the energy of the MM environment calculated with a force field, while E^{QM} is the energy of the QM portion that includes electrostatic and polarization contributions from the MM environment when the electrostatic embedding scheme is used.

LED can be applied to further decompose the QM part of the interaction energy. Let us demonstrate this on the interaction of spin-triplet di(1-adamantyl)carbene with water molecules (see the LEDAW repository for the corresponding ORCA outputs).

4.3.2.1. Supersystem Setup

The input file for the supersystem using HFLD as the QM method is as follows:

```
! UHF HFLD RIJCOSX def2-TZVP(-f) def2/J def2-TZVP/C

%pointcharges "adduct_pc.pc"

%mdci DoLEDhf True end

*xyz 0 3
  C(1)   -1.08752192553253     -0.39785811660581     -1.31255651043615
  C(1)   -0.85484574357832     -1.08485274545435     -0.03049050756303
  C(1)   -0.93184619465148     -0.74815241651238      1.39998574613725
  …
  O(5)   -3.52856434889771     -3.55541023859884     -5.15556558062646
  H(5)   -3.82227421001620     -3.76743656198486     -4.25139907965476
  H(5)   -2.57626609626977     -3.76792799643335     -5.10862423293110
*
```

The coordinates in the xyz block correspond to the QM region of a previous standard QM/MM computation (typically performed with DFT). The point charge file from the previous QM/MM run is read using the `%pointcharges` flag.

If the `*.pc` file is not present in your standard QM/MM directory, it is likely not copied automatically from the scratch directory by your submission script. In that case, you can generate it by running a fast standard single-point QM/MM calculation (the choice of QM method and basis set unimportant) with the following keyword:

```
%qmmm keep_qmmm_file True end
```

For the interaction energy calculations, if fragments are moved from the MM region into the QM region (by adding them to the `xyz` block), the corresponding coordinates must be removed from the `*.pc` file and the first line in this file, corresponding to number of point charges, must be updated. The reverse applies if fragments are shifted from QM to MM.

For this supersystem calculation, the reference QRO energy is printed in the ORCA output file as:

```
E(0)                                          ...   -1117.247592309
```

However, LED part reports the reference energy as:

```
Total REF. energy                      =     -1117.103871349748
```

As in the implicit solvation schemes, this difference arises from the exclusion of the electrostatic interaction energy between QM and MM (point charges) regions from the QM energy in the LED part, while still retaining the polarization of the QM region by the MM region.

The energy reported in the LED part can be reproduced by running a single-iteration UHF calculation on the `*.qro` file obtained from the above supersystem calculation, but without requesting point charge correction:

```
! UHF RIJCOSX def2-TZVP(-f) def2/J MOREAD

%moinp "ADDUCT.qro"

%scf maxiter 1 end

*xyz 0 3
```

Here we read the `*.qro` file as the system is an open-shell system. For closed-shell systems, the `*.qro` and `*.gbw` files are identical. The above calculation will likely crash because it will not converge in one iteration. This is fine—we only need the energy from the first SCF iteration. For this example, the relevant part of the output is as follows:

```
--------------------------------------D-I-I-S--------------------------------------------
Iteration    Energy (Eh)           Delta-E    RMSDP     MaxDP     DIISErr    Damp  Time(sec)
-----------------------------------------------------------------------------------------
               ***  Starting incremental Fock matrix formation  ***
   1    -1117.1037480146019334     0.00e+00  9.27e-04  1.53e-02  3.62e-02  0.700  10.1
```

Note that this energy is consistent with the one reported in the previous LED calculation within 1 mE_h.

4.3.2.2. Subsystem Setup

If a subsystem consists of multiple fragments, its input file follows the same structure as the supersystem, with coordinates, charge, and multiplicity adjusted accordingly:

```
! UHF HFLD RIJCOSX def2-TZVP(-f) def2/J def2-TZVP/C

%pointcharges "adduct_pc.pc"

%mdci DoLEDhf True end

*xyz 0 3
```

Here, since an LED calculation is carried out, the reference energy is printed in the ORCA output both including and excluding QM/MM electrostatics (see the supersystem section).

Now let us consider a single-fragment subsystem. To obtain `E(0)` reference energy (QRO), we setup a cost-effective correlation calculation:

```
! UHF DLPNO-CCSD RIJCOSX def2-TZVP(-f) def2/J def2-tzvp/C LOOSEPNO

%pointcharges "adduct_pc.pc"

%mdci maxiter 0 end

*xyz 0 3
  C   -1.08752192553253     -0.39785811660581     -1.31255651043615
  C   -0.85484574357832     -1.08485274545435     -0.03049050756303
  C   -0.93184619465148     -0.74815241651238      1.39998574613725
  …
*
```

This calculation provides only the QM energy including QM/MM electrostatics. To exclude the electrostatic contribution, a single-iteration UHF calculation without point charges is required, reading the `*.qro` file of the corresponding monomer from the subsystem calculation above. Analogous calculations must be performed for all subsystems.

4.3.2.3. Adjustments for Running LEDAW with QM/MM Outputs

As described above, single-fragment subsystem files lack a necessary energy for performing LED analysis. Therefore, before running LEDAW, this information must be manually added to the ORCA outputs. To maintain a consistent workflow for both single- and multi-fragment systems, and to keep usage simple, LEDAW has not been implemented as QM/MM-aware.

For QM/MM calculations, the recommended approach is to adapt the QM/MM outputs to mimic the implicit solvation outputs, since LEDAW automatically handles the later. Let us demonstrate this adjustment using the supersystem as an example.

Above we reported `E(0)` of the supersystem as `-1117.247592309` E_h and `-1117.103871349748` E_h with and without QM/MM electrostatics, respectively. The later value can come either from the LED output or single-iteration SCF calculation performed as described above. By subtracting these two numbers, the QM/MM electrostatics is found as `-0.143720959` E_h. This value should be added to the supersystem output —preferably at the very bottom—in the following format, as if it were a CPCM dielectric term:

```
CPCM Dielectric    :         -0.143720959 Eh
```

Once such an adjustment is applied to the supersystem and all subsystems, LEDAW can be run on the modified ORCA QM/MM outputs. When this approach is used, the decomposed QM/MM electrostatics will be labeled as `SOLV` within LEDAW.

4.4. LED of Frozen-State and Orbital Relaxation

To illustrate how to dissect the permanent (frozen-state) and induced (orbital relaxation) components of the interaction energy upon adduct formation, we consider water dimer. The interaction energy of the water dimer and its LED components at the DLPNO-CCSD(T_1)/TightPNO/LED level can be computed using the following inputs:

Supersystem:

```
! DLPNO-CCSD(T1) aug-cc-pVTZ RIJK aug-cc-pVQZ/JK aug-cc-pVTZ/C
! VeryTightSCF TightPNO LED

*xyz 0 1
O(1)       -0.702196054    -0.056060256     0.009942262
H(1)       -1.022193224     0.846775782    -0.011488714
H(1)        0.257521062     0.042121496     0.005218999
O(2)        2.220871067     0.026716792     0.000620476
H(2)        2.597492682    -0.411663274     0.766744858
H(2)        2.593135384    -0.449496183    -0.744782026
*
```

Subsystem 1:

```
! DLPNO-CCSD(T1) aug-cc-pVTZ RIJK aug-cc-pVQZ/JK aug-cc-pVTZ/C
! VeryTightSCF TightPNO

*xyz 0 1
O       -0.702196054    -0.056060256     0.009942262
H       -1.022193224     0.846775782    -0.011488714
H        0.257521062     0.042121496     0.005218999
*
```

Subsystem 2:

```
! DLPNO-CCSD(T1) aug-cc-pVTZ RIJK aug-cc-pVQZ/JK aug-cc-pVTZ/C
! VeryTightSCF TightPNO

*xyz 0 1
O        2.220871067     0.026716792     0.000620476
H        2.597492682    -0.411663274     0.766744858
H        2.593135384    -0.449496183    -0.744782026
*
```

Note that, for simplicity, subsystem inputs are given without ghost fragments. However, if BSSE correction is aimed, they can also be included.

In the N-body module of LEDAW, when the outputs of these three ORCA runs are used, LED analysis is performed on both HF and correlation parts, incorporating both permanent and induced interaction components.

Frozen-state LED is implemented in ORCA only for the reference level. To perform this analysis, the antisymmetrized product of the HF wave functions of the isolated fragments (stored, for example, in `frag1.gbw` and `frag2.gbw`) from the last two inputs above is required. This wave function can be obtained by calling the `orca_mergefrag` utility in the command line as follows:

```
orca_mergefrag frag1.gbw frag2.gbw combined_frags.gbw
```

Then, LED must be performed directly on the resulting "`combined_frags.gbw`" wave function. To do this, read the `gbw` file using the `MOREAD` and `moinp` keywords, while avoiding any SCF iterations by using `NOITER` in the simple input line or setting "`maxiter 0`" in the `scf` block. Noting that correlation data is not required but HF/LED can only be requested within a DLPNO calculation, a cost-effective correlation calculation can be set up as described in the preceding sections. Below is a sample supersystem input for obtaining the frozen-state HF/LED data:

```
! HFLD aug-cc-pVTZ RIJK aug-cc-pVQZ/JK aug-cc-pVTZ/C VeryTightSCF LoosePNO

! MOREAD
%moinp "./combined_frags.gbw"

%scf maxiter 0 end

%mdci maxiter 0 DoLedHF True end

*xyz 0 1
O(1)       -0.702196054    -0.056060256     0.009942262
H(1)       -1.022193224     0.846775782    -0.011488714
H(1)        0.257521062     0.042121496     0.005218999
O(2)        2.220871067     0.026716792     0.000620476
H(2)        2.597492682    -0.411663274     0.766744858
H(2)        2.593135384    -0.449496183    -0.744782026
*
```

In this input, it is important to ensure that the coordinates and their order in the `xyz` block match those in the combined `gbw` file obtained with the `orca_mergefrag` utility.

When the output files of this supersystem input and the two subsystem inputs, provided at the beginning of this section, are used in the two-body module of LEDAW, frozen-state HF/LED interaction energy components are obtained, including separate redundant correlation data.

When both N-body and two-body analyses are requested in a single LEDAW-GUI run, cooperativity analysis is automatically performed by subtracting the two-body interaction energy components from the N-body components. Within the context described in this section, this difference at the reference level corresponds to orbital relaxation. Noting that the two-body and cooperativity modules also provide separate correlation data, these must be directly discarded by the user; only the correlation data from the N-body module, representing the fully relaxed state, should be considered. Since all three modules of LEDAW are designed for general-purpose use, redundant correlation data is not excluded automatically in an orbital relaxation run.

For two-fragment systems, the N-body and two-body modules of LEDAW yield identical results when processing the same ORCA output files. This allows for the assessment of differential effects in two-fragment systems by comparing results from these modules under varying computational settings. For instance, users can evaluate the impact of BSSE correction (by performing LED analyses with and without correction) or the influence of the basis set (by performing LED with two different basis sets). LEDAW includes an example for computing BSSE effects on the LED terms of water dimer. However, for multi-fragment systems, analyzing such differential effects via the GUI is not possible, as the two-body module is for a series of two-fragment systems. A script-based workflow, conversely, can readily handle this by calling the N-body engine for two distinct settings and then utilizing the cooperativity engine to evaluate their differences.

4.5. Automating ORCA LED Input File Preparation with LEDIP

As demonstrated above, preparing ORCA LED input files, particularly for multi-fragment systems, can be a time-consuming and error-prone task. To mitigate this, we provide `ledip` (LED Input Preparator), a companion package designed to streamline this process through script-based workflows, facilitating both fragment labeling and ORCA input preparation.

4.5.1. Automatic Fragmentation

4.5.1.1. Sample Usage

The `fragmentation_engine()` function within the `ledip` package is a powerful tool for automatically identifying molecular fragments from an XYZ file. A sample Python script (*e.g.*, `run_fragmentor.py`) for calling `fragmentation_engine()` is shown below:

```
from ledip_package import *

xyzfile = 'test.xyz'

cutoff = 1.62
cut_bonds = [(0, 1), (5, 6), {17, 20}, [32, 35]]
nonstandard_coordination_numbers = {'Bi': 4, '7-9,11,13': 3}

fragmentation_engine(xyzfile, cutoff=cutoff, cut_bonds=cut_bonds,
      nonstandard_coordination_numbers=nonstandard_coordination_numbers)
```

4.5.1.2. Fragmentation Parameters

As seen in the sample input, besides the input XYZ file, the `fragmentation_engine()` accepts three optional user-defined parameters that control how fragments are assigned. These options typically provide sufficient flexibility for fragmenting a supersystem. For more advanced or specific needs, consider leveraging ORCA directly; ORCA 6.1+ offers significantly enhanced built-in fragmentation features.

These parameters can be described as follows:

- **`cutoff`** (*float*, default: 1.8 Å)

  The maximum interatomic distance for two atoms to be considered bonded. The engine constructs an adjacency graph where atoms within this threshold are connected.

  Guidance: Choose a value large enough to capture all true covalent bonds. Typical C–H and C–C bond lengths range from ~1.0 to 1.5 Å, so a value between 1.5 and 2.0 Å is usually suitable. Smaller values may over-fragment the molecule, while larger values risk merging distant atoms.

- **`cut_bonds`** (*list of pairs*, default: `None`)

  A list of 0-based atom index pairs that must not be treated as bonded, even if their interatomic distance falls below the `cutoff`. Each pair can be specified as a list, tuple, or set containing two atom indices. Mixed usage, as shown in the sample input, is supported.

  Use Cases: This parameter is crucial for overriding incorrect automatic bonding (*e.g.*, very strong noncovalent interactions misinterpreted as covalent bonds) or for deliberately breaking existing covalent bonds.

- **`nonstandard_coordination_numbers`** (*dictionary*, default: `None`)

  Defines atoms that must connect to a fixed number of neighboring fragments, overriding the default distance-based bonding rules.

  Use Cases: This is especially useful for handling coordination bonds, where standard distance thresholds may fail to detect bonding between a central atom and its ligands or fail to sever such coordination when needed.

  Mechanism: Internally, the engine uses a union–find structure. It computes fragment centroids and merges fragments based on their proximity to the specified "hub" atoms.

  Dictionary Format: The keys can be either element symbols or explicit 0-based atom index ranges. The values are integers representing the number of fragments each atom must be connected to. For example:

  - `{'Bi': 4}` connects every Bi atom to its four nearest fragments, effectively merging them into a single labeled unit.
  - `{'7-9,11,13': 3}` connects atoms with indices 7, 8, 9, 11, and 13 to three nearest neighboring fragments each.
  - `{'Bi': 0}` ensures that all Bi atoms remain isolated and do not merge with other fragments.
  - **Precedence:** If both element symbols and index-based keys are present, index-based rules take precedence for overlapping atoms.

4.5.1.3. Input XYZ File Format

A valid XYZ file must begin with a line specifying the number of atoms, followed by a mandatory comment line. Each subsequent line must contain an element symbol followed by its space-separated X, Y, Z coordinates. The engine also recognizes optional fragment labels appended after the element symbol in parentheses for some or all atoms (*e.g.,* C(3), O(2)). Labels with extra whitespace (*e.g.,* `C ( 3 )`) are also correctly interpreted.

4.5.1.4. Fragmentation Engine Behavior Modes

The engine behavior depends on whether the input XYZ file includes fragment labels:

- **Automatic Fragmentation Using Unlabeled or Partially Labeled Input XYZ File**

  If the input XYZ file (*e.g.*, `test.xyz`) does not contain any fragment labels (or only some atoms are labeled), the engine automatically assigns each unlabeled atom to a fragment based on the three user-defined parameters. The labeled structure is written to a new file (*e.g.*, `test_labeled.xyz`).

  The atomic coordinates are reordered so that atoms belonging to the same fragment appear consecutively. Fragments are labeled sequentially in the order of detection, starting from 1, unless those labels are already present in the input XYZ file. For example, the first atom read is assigned to fragment 1; if the next atom belongs to a different fragment, it is assigned to fragment 2, and so on. If label 2 is already used in the input XYZ file, the next available label (e.g., 3) is assigned instead. Fragment labels are appended to the element symbols in parentheses.

  The engine prints a summary to the console, such as:

```
Unlabeled atoms will be automatically fragmented. Existing fragment
labels (if any) will be preserved.

Processing 462 unlabeled atoms...

16 new fragments generated from unlabeled atoms.

Updated cut_bonds (mapped to new final atom indices):
[[0, 135], [1, 149], [2, 164], [3, 179]]

Updated nonstandard_coordination_numbers (keys mapped to new final atom
indices):
{'Bi': 4, '11,14,16,21-23': 3}

462 atoms processed.

Output written to test_labeled.xyz

Fragmentation engine completed successfully!
```

  This summary includes the number of atoms processed, the number of fragments generated, and updated `cut_bonds` and `nonstandard_coordination_numbers` according to the new atom ordering. It is essential to use these updated values in the Python input file if you plan to further process the `test_labeled.xyz` file (*e.g.*, in subsequent `fragmentation_engine()` runs or for ORCA input).

  If actual covalent bonds are cut, and thus the CovaLED scheme is going to be used, the updated `cut_bonds` pairs must also be included in the ORCA input file:

```
%mdci covalent {0 135} {1 149} {2 164} {3 179} end
```

- **Refining Fragments**

  If the initial fragmentation results do not fully meet your expectations, you can selectively reprocess only the incorrect portions. To do this:

  (*i*) Remove labels from incorrect fragments while retaining those of the correctly identified ones.

  (*ii*) Update `cut_bonds` and `nonstandard_coordination_numbers` entries from the console output from the previous run as well as the XYZ filename.

  (*iii*) Rerun the engine using the modified XYZ file and revised parameters.

  In this mode, the engine preserves existing labels and processes only the unlabeled atoms.

  New fragments generated from unlabeled atoms are assigned labels using a gap-filling strategy. Hence, the lowest unused label numbers are filled first to maintain sequential labeling.

  Keep in mind that atomic coordinates are reordered (atoms of the same fragment appear consecutively; fragments sorted by label). As a result, atom indices may change, so it is critical to always use the updated values printed in the console after each run to ensure consistency.

- **Reordering Fragment Labels**

  If the input XYZ file already includes fragment labels for all atoms, the engine will reorder atoms as follows:

  Atom Reordering: Atoms belonging to the same fragment appear consecutively.

  Sequential Relabeling: Fragments are relabeled sequentially from 1 based on their order of appearance in the input XYZ file, preserving their composition. For example, if the input XYZ file begins with fragments labeled 9, 7, and 4, they are relabeled as 1, 2, and 3 respectively. The relabeled structure is written to a new file (*e.g.*, `test_relabeled.xyz`).

  Controlling Order: To control this new sequential label order, simply move at least one atom from each desired fragment to the top of the input XYZ file in the desired order.

  Parameter Ignoring: The `cut_bonds` and `nonstandard_coordination_numbers` parameters are ignored in this mode, as all fragment compositions are already explicitly defined in the input XYZ file.

  Note on Relabeling: If you choose to preserve the original labels at this stage, they can still be reordered using LEDAW's relabeling feature while processing the LED data.

4.5.2. Automatic Input File Generation

4.5.2.1. Sample Usage

The `led_input_prep_engine()` function within the `ledip` package automates the generation of ORCA input files required for N-body and two-body LED calculations, with and without BSSE correction. This streamlines what would otherwise be a tedious and error-prone manual task, especially for systems composed of many fragments.

A sample Python script (*e.g.*, `run_inp_prep.py`) for calling `led_input_prep_engine()` is shown below:

```
from ledip_package import *

xyzfile = 'test_labeled.xyz'

output_dir = 'LED-INPS/DNA'

subsystems = [[1,9],[2,3,4,5,6,7,8],[10,11,12,13,14,15,16]]

orca_inp_heading = '''
! DLPNO-CCSD(T) RIJCOSX def2-tzvp(-f) def2/J def2-tzvp/C VeryTightSCF
! CPCM(water) DefGrid3 NormalPNO LED

%pal nprocs 16 end
%maxcore 6000
%scf maxiter 999 end

%mdci TCutPairs 1e-5 end

*xyz 0 1
'''

led_input_prep_engine(xyzfile, subsystems, orca_inp_heading, output_dir)
```

4.5.2.2. Input Preparation Parameters

The `led_input_prep_engine()` function uses the following parameters to tailor the generated ORCA input files:

- **`xyzfile`** (*string*)

  The path to the labeled XYZ coordinate file of the supersystem. This file can be the output from `fragmentation_engine()` or an external source. The `led_input_prep_engine()` function reads fragment definitions (labels and coordinates) from this file.

- **`subsystems`** (*list of lists of integers*)

  A list of subsystems, where each inner list defines a subsystem in terms of its fragment labels in the `xyzfile`.

  Example: `[[1,9], [2,3,4,5,6,7,8], [10,11,12,13,14,15,16]]`

  - In a DNA duplex example, `[1,9]` might correspond to the labels of two backbones (subsystem 1). Note that `[1]` and `[9]` could also be set to represent each backbone as a separate subsystem.
  - `[2,3,4,5,6,7,8]` could be the labels of nucleobases on strand 1 (subsystem 2).
  - `[10,11,12,13,14,15,16]` could be the labels of nucleobases on strand 2 (subsystem 3).

  Empty List Option: If `subsystems = []` (an empty list), then each fragment in the `xyzfile` file is automatically treated as a separate subsystem.

  - **`orca_inp_heading`** (*multi-line string*)

  This string contains the complete ORCA input block that will be placed at the beginning of all ORCA input files. Ensure the last line of this string includes the `*xyz` along with charge and multiplicity (*e.g.*, `*xyz 0 1`). The engine appends the corresponding coordinate lines from the labeled `xyzfile` file after this heading in the ORCA input files.

  Automatic Heading Adjustments For Single-Fragment Subsystems: For subsystems containing only a single *real* fragment (excluding ghost fragments), the engine automatically adjusts the *orca_inp_heading* as follows:

  - Always removed:
    - The `LED` keyword in the simple input line (`! ... LED`)
    - The `DoLEDHF` directive within the `mdci` block.
  - Additionally removed if the calculation method is `HFLD`:
    - The `HFLD` keyword itself.
    - The PNO setting keywords (`LoosePNO`, `NormalPNO`, `TightPNO`).
    - The entire `mdci` block.
- **`output_dir (string)`**

  The path to the directory where all the generated ORCA input files will be saved. The engine will create this directory if it doesn't already exist.

  Example: `output_dir = 'LED-INPS/DNA'`

  - If the specified `output_dir` path doesn't exist, it will be created.
  - If this path already exists and contains `NBODY` and `TWOBODY` subdirectories, these two directories will be deleted first before new files are written. This prevents mixing old and new input files.

4.5.2.3. Generated Output Structure

Upon successful execution, the engine creates a structured output within the specified `output_dir`.

- **`NBODY directory:`**
  - `supersystem.inp` – Contains full system input
  - `SUBSYS_withBSSE` directory – Contains subsystem inputs including ghost fragments
  - `SUBSYS_woBSSE` directory – Contains subsystem inputs without ghost fragments
- **`TWOBODY directory:`**
  - `DIMER` directory – Contains inputs for all pairs of fragments residing in different subsystems
  - `ONEBODY_withBSSE` directory – Contains two monomer inputs for each fragment pair (one ghost fragment and one real fragment in each).
  - `ONEBODY_woBSSE` directory – Contains monomer inputs for each fragment with no ghost fragments.

4.5.2.4. Important Considerations for Generated Files

- **System Specific Specifications**

  For simplicity, the engine supports only a unique ORCA input file heading. The user may need file-specific modifications, such as setting looser HFLD correlation settings for multi-fragment subsystems (as only HF/LED decomposition is needed), or different `HFFragInter` and `covalent` directives in the `mdci` block across input files, etc. However, these require modifications on only a limited number of input files. Therefore, we encourage the user to apply such settings manually or using command-line tools.
- **Caution for Charge and Multiplicity**

  For simplicity, the engine supports only a single set of charge and multiplicity for all generated input files. However, one can easily adjust these in bulk after input file generation using command-line tools.

  Example: The generated `ONEBODY_withBSSE` directory contains files like `dimerX-Y_X.inp` and `dimerX-Y_Y.inp`. The suffix `_X` or `_Y` indicates the real fragment for which charge and multiplicity might need adjustment.

To replace `*xyz 0 1` with *`xyz -1 3` in all `.inp` files where fragment 4 is the real fragment (`dimer*-*_4.inp`) within that directory, navigate to the directory in your terminal and use `sed`:

```
sed -i 's/*xyz 0 1/*xyz -1 3/g' *_4.inp
```

As this method is generally more practical and less error-prone than managing individual charge and multiplicity lists for all fragments within the Python script, no built-in feature for individual charge/multiplicity replacement is implemented.

- **Redundant Input Files**

  For a single-fragment subsystem without BSSE correction, the generated input files are identical in the `NBODY/SUBSYS_woBSSE` and `TWOBODY/ONEBODY_woBSSE` directories. For completeness, the engine provides the same file in both locations, but with different names consistent with the respective directory's naming convention.

5

LEDAW-GUI

LEDAW is a program package designed to automate and streamline the LED analysis of interaction energies at local coupled-cluster levels within the DLPNO framework. It facilitates the post-processing of ORCA LED output files using both standard LED and fp-LED schemes, with optional CBS and CPS extrapolations. LEDAW also enables the investigation of many-body effects in complex molecular systems. The program generates detailed interaction energy matrices and heat maps, offering in-depth insights into how specific fragment pairs contribute to the overall stability of molecular systems. This chapter explains how to use the `ledaw` package via its GUI and outlines the resulting output file structure.

5.1. Using LEDAW-GUI

5.1.1. Main Window

When LEDAW-GUI is launched, the main window opens to the “Home” tab, as shown in Figure 5.1.

In addition to the “Home” tab, the main window includes the 'Manage Session' and 'About' tabs. The 'Manage Session' tab offers three options:

- "Resume" – Returns to the last visited tab.
- "Start Over" – Restores the default settings at startup, effectively starting a fresh LEDAW session.
- "Exit" – Closes the LEDAW session, equivalent to pressing the “X” button in the top-right corner.

The "About" tab provides citation information and a brief license statement, directing users to the LEDAW repository or its manual for the full version.

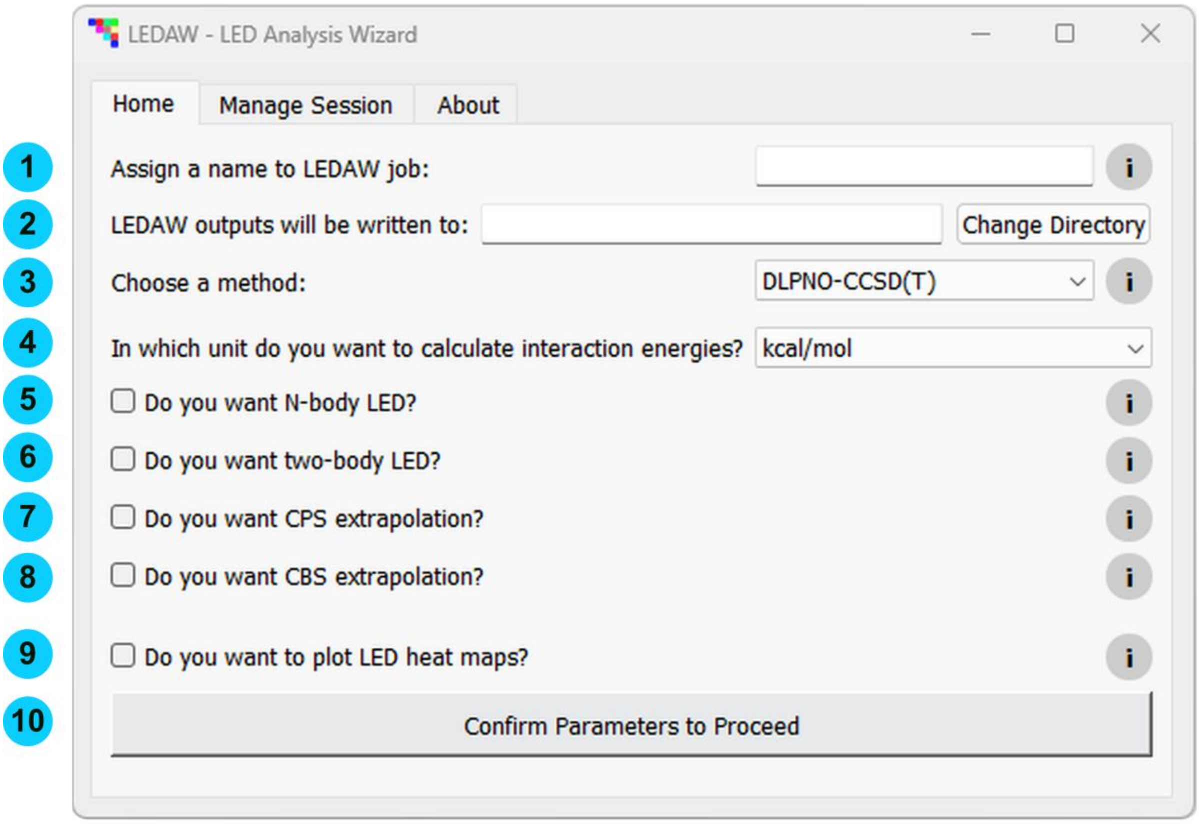


Figure 5.1. Main window of LEDAW-GUI, showing the "Home" tab displayed by default upon startup.

5.1.2. Home Tab

The ten fields in the "Home" tab are labeled on the left side of Figure 5.1. Fields typically include an info button i, a self-explanatory dropdown menu, or confirmation button. Most of the explanations provided in this section can also be accessed interactively by clicking the i buttons. The following describes each field in detail:

1 **Job Name** – This compulsory field prompts you to specify a job name, such as "`WATER-HEXAMER`". Unless a custom path is specified Field 2, LEDAW results will be saved in a "`./LEDAW-OUT/JOBNAME`" directory, where "`JOBNAME`" corresponds to the text entered in this field (*e.g.*, "`WATER-HEXAMER`"), and "`./`" refers to the directory from which LEDAW-GUI is launched.

2 **Output Directory** – This field displays the LEDAW output directory, which is automatically set based on the job name provided in Field 1. You can change this path by typing or pasting a new one into the text box, or navigating to your desired location using the "`Change Directory`" button. If the specified directory does not already exist, it will be generated automatically.

If a custom path is entered in this field but the job name in Field 1 is later changed, the output path in this field will revert to the new default based on the updated job name, overwriting the user-defined path.

If the specified output directory already exists (whether default or user-specified), LEDAW will overwrite its contents. Any existing files or subdirectories in that location may interfere with the run. To avoid potential conflicts, it is strongly recommended to use a unique LEDAW output directory name that does not match any existing subdirectory under the LEDAW output root directory.

③ **Method** – This dropdown field allows you to select one of the three available analysis methods: DLPNO-CCSD(T), DLPNO-CCSD, or HFLD. This selection determines how LED components will be collected from the ORCA output files.

If HFLD is selected, an additional option appears allowing you to use the SCF energy instead of the `E(0)` reference energy. This option should be ticked only for closed-shell systems where at least one subsystem consists of a single fragment and only an RHF (without correlation) calculation was performed on that subsystem. Do not use this option for open-shell systems, as the `E(0)` energy in HFLD/LED analysis is based on QROs, not UHF, making SCF-based substitution invalid.

Note that the method selected here does not have to match the method used in ORCA runs. For instance, even if the ORCA output indicates "DLPNO-CCSD," the correct choice here might be HFLD if the multilevel DLPNO approach was used.

As another example, suppose the ORCA outputs come from a DLPNO-CCSD(T) calculation. If the user selects DLPNO-CCSD, the triples contributions will be ignored. Then, to access the effect of triples contribution, a separate LEDAW run using DLPNO-CCSD(T) should be performed as well.

If HFLD is selected for DLPNO-CCSD or DLPNO-CCSD(T) ORCA outputs, LEDAW will extract only the reference energy terms and inter-subsystem dispersion terms at the DLPNO-CCSD level, corresponding to the target HFLD interaction energy. HFLD is specifically designed to approximate these values.

However, if ORCA outputs belong to HFLD, but DLPNO-CCSD or DLPNO-CCSD(T) is mistakenly selected here, the resulting correlation energies will be meaningless. This is because HFLD outputs contain redundant correlation terms that are used solely to isolate dispersion contributions efficiently. These terms are not physically interpretable on their own, as other correlation components are deliberately omitted in the HFLD framework.

④ **Energy Unit** – This dropdown field allows you to choose the unit for presenting the LED analysis results, either in kcal/mol or kJ/mol.

⑤ **N-body LED** – Tick the checkbox in this optional field to request an N-body LED analysis. This analysis requires ORCA output files for both the supersystem and its corresponding subsystems.

⑥ **Two-body LED** – Tick the checkbox in this optional field to request a two-body LED analysis. This analysis requires ORCA output files for all fragment pairs in a supersystem where the fragments belong to different subsystems, along with the corresponding one-body ORCA outputs for each fragment.

In the p-LED framework, this module is designed to perform multiple two-fragment LED analyses (where at least one fragment in each fragment pair has different coordinates than in another pair) and compiles the results into a matrix. The standard LED aggregates the diagonal elements corresponding to the same fragment across different fragment pairs.

If both N-body LED and two-body LED are selected, cooperativity analysis will be performed automatically.

⑦ **CPS Extrapolation** – Tick the checkbox in this optional field to request CPS extrapolation for the correlation terms in the N-body and/or two-body LED analysis. To enable the CPS extrapolation, ORCA LED output files must be available for each selected LED analysis type obtained using two T_{CutPNO} thresholds with adjacent exponents (*e.g.*, 10^{-6} and 10^{-7}).

⑧ **CBS Extrapolation** – Tick the checkbox in this optional field to request CBS extrapolation for the N-body and/or two-body LED analysis. To enable the CBS extrapolation, ORCA LED output files must be available for each selected LED analysis type, generated using two basis sets with adjacent cardinal numbers (*e.g.*, cc-pVTZ and cc-pVQZ).

When this checkbox is ticked, a dropdown menu appears to select the CBS extrapolation method:

- **CBS(2/3)** – CBS extrapolation using (aug-)cc-pVDZ and (aug-)cc-pVTZ
- **CBS(3/4)** – CBS extrapolation using (aug-)cc-pVTZ and (aug-)cc-pVQZ
- **Other** – CBS extrapolation using custom F_{ref} and F_C extrapolation coefficients. For example, if def2-*n*VP basis sets are used in the ORCA calculations, the corresponding coefficients can be taken from Table 1.1.

⑨ **Plot** – Tick the checkbox in this optional field to request LED interaction energy heat maps based on all the selected settings. To generate the plots, LED summary Excel files (“`Summary_Standard_LED_matrices.xlsx`” and/or “`Summary_fp-LED_matrices.xlsx`”) must exist in subdirectories of the LEDAW output directory specified in Field ②. These Excel files may be generated during the current LEDAW run or, if neither N-body nor two-body LED analysis is selected, may originate from a previous run.

In the latter case, *i.e.*, if only the Plot option is selected (along with a LEDAW output directory), LEDAW-GUI can also serve as a standalone upper-triangle heat map plotter, provided that any directory inside the LEDAW output directory contains Excel files that are named and formatted exactly like a standard LEDAW summary file..

⑩ **Confirmation Box** – To proceed with your selected settings from fields ①-⑨, you must tick at least one of the checkboxes in Field ⑤ (N-body LED), Field ⑥ (two-body LED), or Field ⑨ (Plot), and ensure that a job name and a LEDAW output directory are present in Field ① and Field ②. Once these conditions are met, click the confirmation box in Field ⑩ to review and apply the settings.

Clicking the confirmation box opens a new window listing all your selections. Review this list carefully. To make further changes, press “Cancel” to return to the “Home” tab. Otherwise, press “OK” to proceed. After confirmation, the “Home” tab becomes locked, and no further edits will be allowed.

Depending on the selected options, the following tabs will be generated in sequence: “N-body,” “Two-body,” “Plot,” and “Run.” The content within each tab will dynamically adjust based on your selections, *i.e.*, whether N-body and two-body LED analyses are requested individually or together, and whether CPS and/or CBS extrapolations are also requested.

5.1.3. Basic N-body Tab

When CPS and CBS extrapolations are not requested, the layout of the “N-body” tab appears as shown in Figure 5.2.

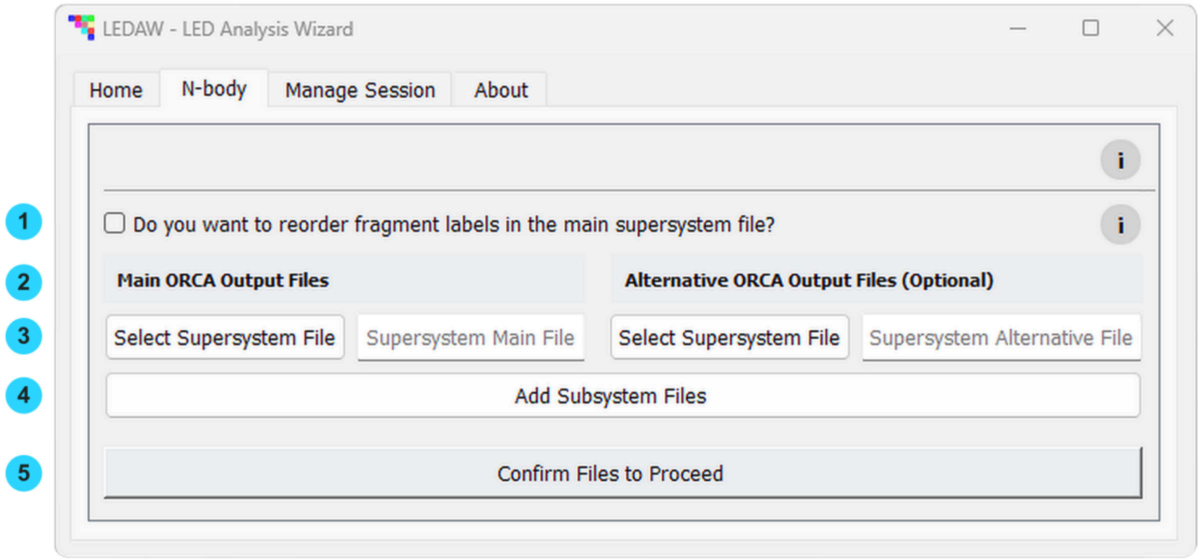


Figure 5.2. The "N-body" tab of the LEDAW-GUI as it appears when N-body LED and/or two-body LED is selected without CPS and CBS extrapolations in the "Home" tab.

The five fields in the basic "N-body" tab are labeled on the left side of Figure 5.2. Most of the explanations provided below, except those related to file selection, can also be accessed interactively by clicking the ⓘ buttons. The following describes each field in detail:

① **Relabel Mapping** – LEDAW inherits fragment labels from the "main supersystem" ORCA output file (see ② below for the distinction between "main" and "alternative"). Fragments in the alternative supersystem file and all subsystem ORCA output files are aligned with those in the main supersystem file by matching their coordinates. However, users may still wish to present the LED analysis results using a different fragment numbering scheme—one that better reflects the system's structural or chemical features.

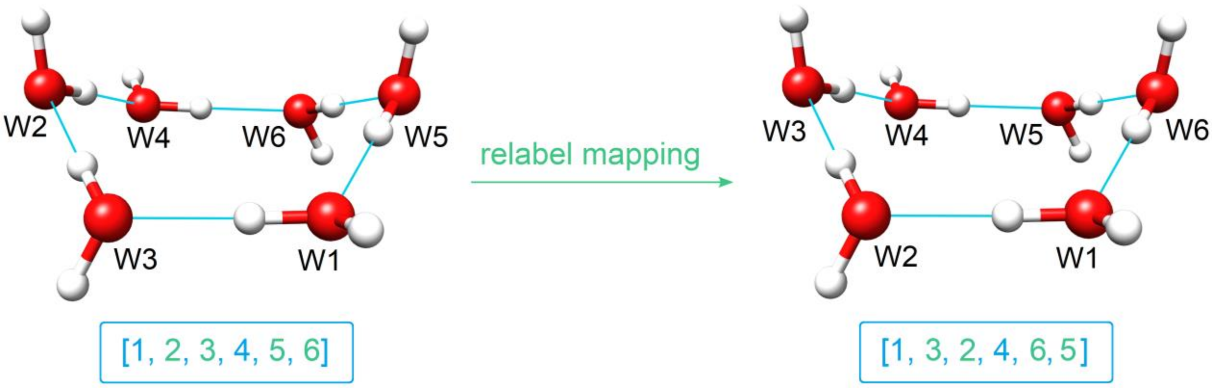


Figure 5.3. (*Left*) Original fragment labeling in the supersystem ORCA output file. (*Right*) Desired fragment labeling scheme, which can be achieved by swapping labels 2 and 3, and labels 5 and 6. Thus, the original ordered label list [1, 2, 3, 4, 5, 6] is mapped to [1, 3, 2, 4, 6, 5]. The latter list is referred to in LEDAW as the "relabel mapping".

For example, the supersystem ORCA output files for the boat conformer of the water hexamer in the LEDAW repository use a fragment labeling scheme shown on the left side of Figure 5.3. When labels 2 and 3, and 5 and 6 are swapped, the resulting labeling scheme better reflects the hydrogen-bond network, making the analysis easier to interpret. Starting from the original ordered list [1, 2, 3, 4, 5, 6], the desired relabeling will then be [1, 3, 2, 4, 6, 5]. This new list is referred to in LEDAW as the "relabel mapping".

The "relabel mapping" list must include all integers from 1 up to the maximum fragment label in the main supersystem file. Note that this maximum label is not necessarily equal to the number of fragments when user-defined fragment labels are non-sequential—a setup that, while not recommended, is still recognized by both ORCA and LEDAW.

For instance, if a user assigns labels [3, 4, 7, 9, 11] to a five-fragment system, ORCA still reserves placeholders for the missing labels [1, 2, 5, 6, 8, 10]. As a result, the full original label list is [1, 2, **3**, **4**, 5, 6, **7**, 8, **9**, 10, **11**], with the actual fragment labels shown in bold. Suppose the user wants to relabel fragments 11, 4, 9, 7, and 3 as 1 through 5, respectively, using the relabeling feature in LEDAW. Then, the complete relabel mapping becomes [**11**, **4**, **9**, **7**, **3**, 1, 2, 5, 6, 8, 10]. The positions assigned to placeholder labels can be filled arbitrarily among themselves, as these entries contain no energy data in the ORCA output and are discarded during the final stage of LEDAW processing. However, care must be taken to preserve the intended mapping of actual fragment labels.

When the checkbox in Field ① is ticked, a textbox appears for the user to enter the relabel mapping as a comma-separated list, with optional use of square brackets. If the mapping doesn't start from 1 or is not sequential, a warning message will appear. If the user does not modify the list after the warning, it will be ignored. Note that no checks are performed on the length of the list. Therefore, if the maximum label is missing or the list size exceeds the maximum label, the LEDAW run will crash in the "Run" tab.

② **Title for Main and Alternative File Blocks** – The file selection section of the "N-body" tab consists of two adjacent blocks: one on the right for "main" ORCA output files and one on the left for "alternative" ORCA output files. Field ② provides title for these two sections.

This design accommodates scenarios where large molecular systems cause ORCA calculations to crash—often due to insufficient memory allocations or storage limitations. In such cases, users may prefer to reuse parts of the output from the failed job rather than rerun the entire calculation. LEDAW allows each system (supersystem and subsystems) to be assigned two ORCA output files: a "main" and an optional "alternative" file. LEDAW first attempts to extract all necessary data from the main file. If certain data are missing, it automatically checks the corresponding alternative file. When used, both files for a given system must be listed on the same line.

This dual-file feature supports several practical scenarios. For example:

- If an ORCA job crashes after completing the HF/LED analysis but before the correlation decomposition, the HF/LED step can be skipped in the resubmission to save time. LEDAW can then retrieve HF/LED data from the original run and correlation data from the resubmitted job.
- If HF/LED analysis was unintentionally omitted in an HFLD calculation, the user may choose to rerun the ORCA job with looser correlation settings to quickly obtain the missing HF/LED data.
- A user may initially perform a DLPNO-CCSD/LED calculation and later decide to rerun the analysis at the DLPNO-CCSD(T)/LED level. In this case, HF/LED can be

deactivated in the second run for efficiency, as it is already available from the previous calculation.

③ **Select Supersystem File** – To select the main supersystem file, click the "Select Supersystem File" button under the "Main ORCA Output Files" block. In the pop-up window, navigate to the directory containing the desired ORCA output file, select the appropriate file, and confirm your selection.

To specify an optional alternative supersystem file, use the corresponding button under the "Alternative ORCA Output Files (Optional)" block on the left. Alternatively, you can manually enter or paste file paths into the text fields next to each button.

④ **Add Subsystem Files** – To select the main subsystem files, click the "Add Subsystem Files" button. In the pop-up window, navigate to the directory containing the desired ORCA output files, select all relevant files, and confirm your selection. A new line will be created for each selected subsystem, with the main file path populated and the alternative field left empty. If the subsystem ORCA files are located in different directories or if any were missed in the initial selection, you can repeat this process multiple times by pressing the "Add Subsystem Files" button again.

To specify an alternative file for a subsystem, click the "Select Subsystem File" button in the alternative file section on the right, aligned with the corresponding subsystem. Manual entry or pasting of file paths is also supported in all associated text fields.

⑤ **Confirmation Box** – Clicking the "Confirm Files to Proceed" button opens a dialog window listing all your selections. Review this list carefully. To make further changes, press "Cancel" to return to the "N-body" tab. Otherwise, press "OK" to proceed. After confirmation, the "N-body" tab becomes locked, and no further edits will be allowed. If neither a two-body LED nor a plot job is requested, this action will also create a "Run" tab that displays the LEDAW run status and automatically initiates the LEDAW run.

LEDAW-GUI does not read the contents of the specified ORCA output files until file selections are confirmed in the last dynamically generated tab. Therefore, it is the user's responsibility to ensure that all provided files are valid and appropriate. Incorrect file specifications may result in application crashes or produce unreliable results.

During data collection, LEDAW-GUI performs only basic validation checks and displays warning popups. These involve verifying the presence of a supersystem file, ensuring that at least two subsystem files are provided, detecting inconsistencies in the number of files across additional sections of the "N-body" and "Two-body" tabs when CPS and/or CBS extrapolation is enabled, and flagging clearly invalid relabel mappings (e.g., mappings that are not sequential or do not start from 1). However, no checks are performed on the length of the relabel mapping list, as already discussed above.

5.1.4. Basic Two-body Tab

When CPS and CBS extrapolations are not requested, the layout of the "Two-body" tab appears as shown in Figure 5.4. This basic layout includes two section headings: "One-body ORCA Output Files" and "Two-body ORCA Output Files". A 'Select Files' button is located below each section heading.

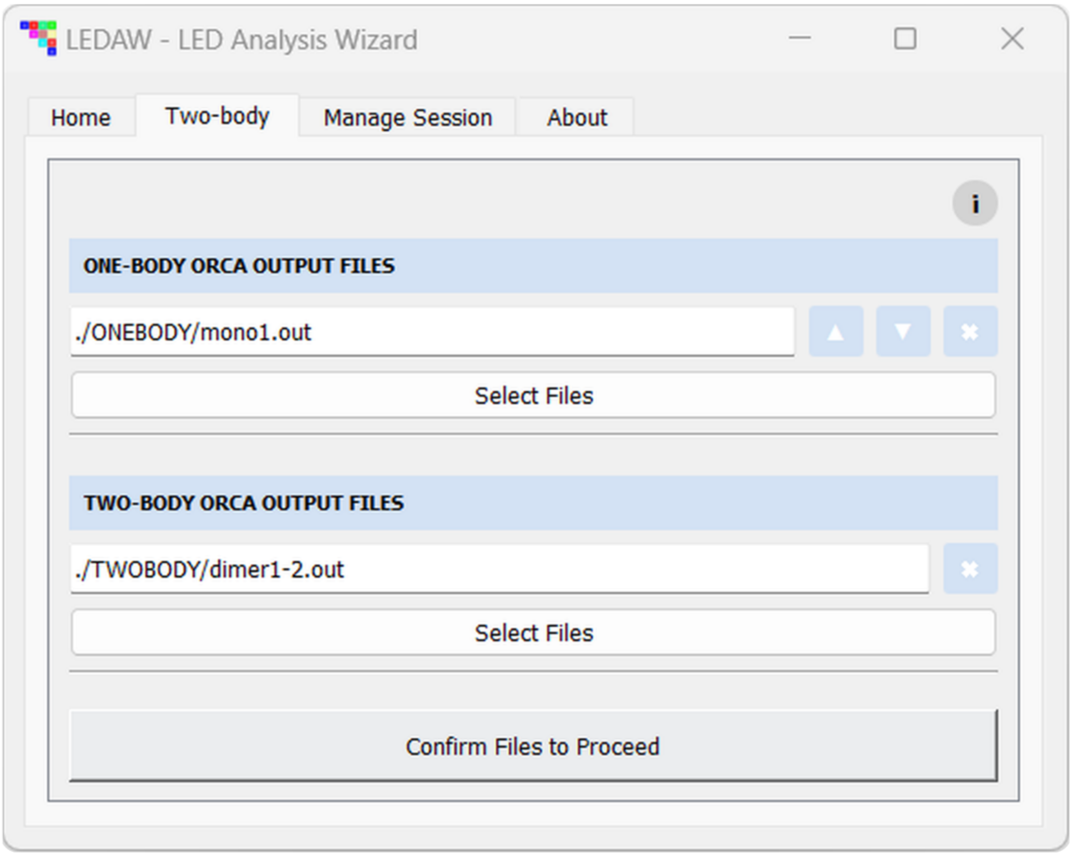


Figure 5.4. The "Two-body" tab of the LEDAW-GUI as it appears when two-body LED is selected, either alone or together with N-body LED, without CPS and/or CBS extrapolation in the "Home" tab. The view shown corresponds to the state after one one-body and one two-body ORCA output file have been selected.

To add the one-body and two-body ORCA output files to the layout, click the corresponding "Select Files" button. In the pop-up window, navigate to the directory containing the desired ORCA output files, select all relevant files, and confirm your selection. A new line will be added for each selected file under the corresponding section heading. If the ORCA files are located in different directories or if any were missed during the initial selection, you can repeat this process as needed by pressing the "Select Files" button again. Each file path can also be manually edited or replaced by pasting a new path. It is worth noting again that only two-body files involving fragments from different subsystems must be included in the two-body analysis.

Figure 5.4 shows the layout after one one-body and one two-body ORCA output file have been added. The one-body line includes up (↑), down (↓), and "X" buttons on the right, allowing the user to reorder the list or remove the entry. The two-body line includes only the "X" button, which removes the corresponding line. Unlike in the N-body case, alternative file selection is not supported as two-body LED computations are already relatively inexpensive.

If both N-body LED and two-body LED are selected in the "Home" tab, the fragment labeling used in the N-body analysis is automatically inherited by the two-body analysis to ensure consistency and facilitate comparison. However, if N-body LED is not selected, LEDAW assigns fragment labels from 1 up to the total number of fragments in the supersystem based on the order in which the one-body ORCA output files are listed in the "One-body ORCA Output Files" section.

Without BSSE correction, only one one-body file per fragment is required. Fragment labels are assigned in the order the one-body files appear if N-body LED is not selected. For example, if the one-body section lists frag3.out, frag1.out, and frag2.out in that order, the fragments will be labeled as 1 (frag3), 2 (frag1), and 3 (frag2), respectively.

With BSSE correction, two one-body files are needed for each two-body file—one for each fragment in the dimer, with the other treated as a ghost fragment. Let us consider all three dimers of a three-fragment supersystem,: 1–2, 1–3, and 2–3. Each dimer requires two one-body files, one per real fragment. These files can be named following a "dimerX-Y_Z" convention, where "X-Y" identifies the dimer and "Z" specifies the real fragment (either X or Y), while the other fragment is treated as a ghost.

Suppose the "One-body ORCA Output Files" section includes the following files in order: dimer2-3_3, dimer1-2_1, dimer1-3_3, dimer1-3_1, dimer2-3_2, dimer1-2_2. These correspond to real fragment **3**, **1**, 3, 1, **2**, 2, respectively. If N-body LED is not selected, LEDAW assigns internal labels based on the first occurrence of each unique real fragment in this list, regardless of which dimer file it comes from. In this case, skipping repeated labels, the user-specified fragment ordering is 3, 1, 2, which LEDAW assigns internal labels 1, 2, and 3, respectively. Hence, fragment 3 becomes 1, fragment 1 becomes 2, and fragment 2 becomes 3. If this is not your intended labeling scheme, use the up/down buttons or manually edit the file paths to adjust the order of the fragments as needed.

After selecting all one-body and two-body files, click the "Confirm Files to Proceed" button. This opens a dialog window listing all your selections. Review this list carefully. To make further changes, press "Cancel" to return to the "Two-body" tab. Otherwise, press "OK" to proceed. After confirmation, the "Two-body" tab becomes locked, and no further edits will be allowed. If plot job is not requested, this action will also create a "Run" tab that displays the LEDAW run status and automatically initiates the LEDAW run.

Note that if both N-body LED and two-body LED are requested without BSSE correction, and each subsystem consists of a single fragment, then the subsystem files specified in the 'N-body' tab and the one-body files specified in the 'Two-body' tab must be identical.

5.1.5. N-body and Two-body Tabs with CPS and/or CBS Extrapolation

When CPS and/or CBS extrapolations are requested, the layout blocks ①–④ in the 'N-body' tab (as shown in Figure 5.2) are duplicated, creating multiple sections within the tab. Similarly, in the "Two-body" tab, the "Select Files" button under both the one-body and two-body sections is replicated for each computational setting. Each section in both the "N-body" and "Two-body" tabs is clearly labeled at their top, corresponding to the respective computational setting, as follows:

- **Only with CPS** – Looser PNO; Tighter PNO
- **Only with CBS** – Smaller Basis Set; Larger Basis Set
- **Both with CPS and CBS** –
  - Smaller Basis Set and Looser PNO
  - Smaller Basis Set and Tighter PNO
  - Larger Basis Set and Looser PNO
  - Larger Basis Set and Tighter PNO

Each of these sections allows users to specify the appropriate ORCA output files for the corresponding computational setting. In the "N-body" tab, the relabel mapping applies only within the corresponding section and can therefore be adjusted for each individual main supersystem file.

As discussed earlier, “smaller” and “larger” basis sets refer to those with adjacent cardinal numbers (*e.g.*, cc-pVQZ and cc-pV5Z, respectively), required for CBS extrapolation. Likewise, “looser” and “tighter” PNO settings refer to T_{CutPNO} values with adjacent exponents (*e.g.*, 10^{-7} and 10^{-8}, respectively), required for CPS extrapolation.

The “N-body” and “Two-body” tabs each include only one “Confirm Files to Proceed” button, which allows users to review and confirm all settings at once. After confirmation, the corresponding tabs are locked, and no further edits can be made.

5.1.6. Plot Tab

The layout of the 'Plot' tab is the same irrespective of the selected analysis or extrapolation settings (see Figure 5.5). It consists of three blocks. The first block lists adjustable parameters for designing the heat maps. The second and third blocks allow the user to enter corresponding values for standard LED and fp-LED, respectively. These input fields are pre-populated with default values but can be modified as desired. The parameters are explained below:

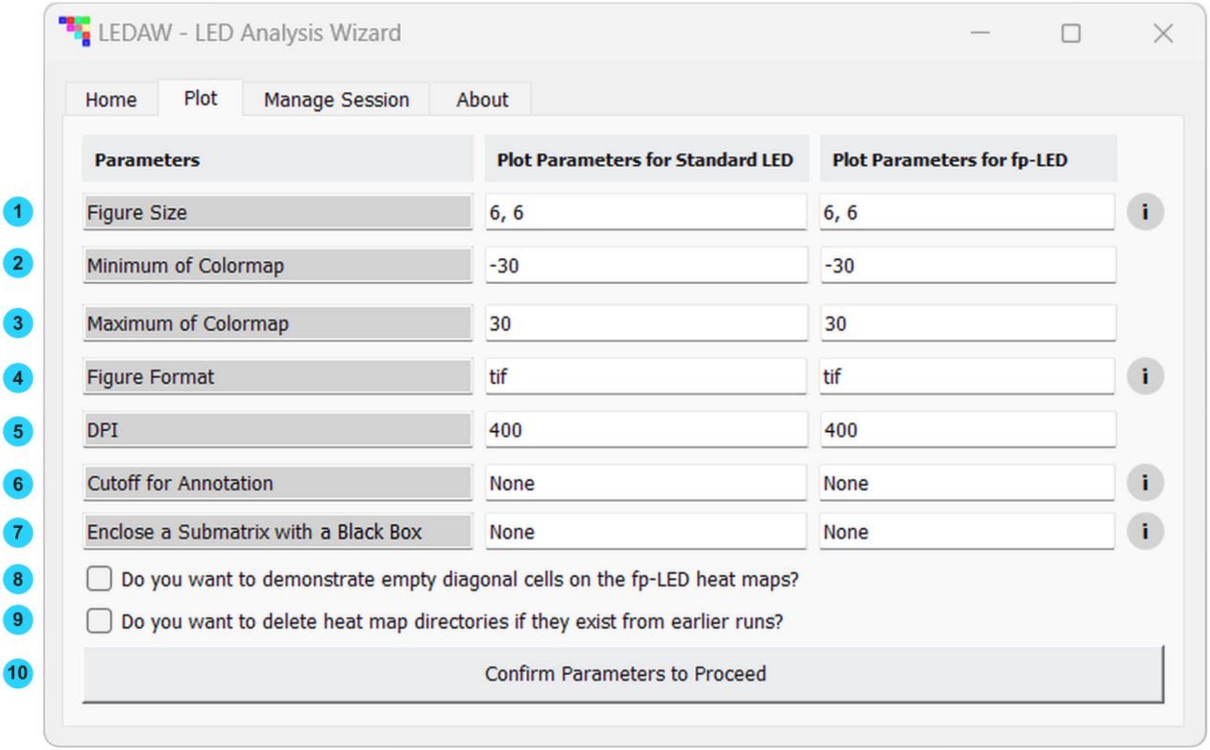


Figure 5.5. The “Plot” tab of the LEDAW-GUI.

① **Figure Size** – Specify the size of each heat map as comma-separated float numbers for “width” and “height”, respectively. For a good balance between the sizes of annotated values and cells, setting both values equal to the number of fragments in the system typically yields optimal results.

② **Minimum of Color Map** – Set approximately equal to the largest negative value in the interaction energy matrices. Any smaller value will be clamped to the color representing this minimum.

③ **Maximum of Color Map** – Set approximately equal to the largest positive value in the interaction energy matrices. Any higher value will be clamped to the color representing this maximum.

④ **Figure Format** – Select from a list of supported image formats, listed when click to the ⓘ info button. For post-processing or storage efficiency, a vector format is recommended.

⑤ **DPI** – Adjust the resolution of the heat map figures in DPI.

⑥ **Cutoff for Annotation** – To annotate only significant values, specify a cutoff value (*e.g.*, X = 0.01). Values between –X and +X will not be labeled. If left as "None," all values are annotated.

⑦ **Enclose a Submatrix with a Black Box** – To highlight specific interactions terms, such as genuine inter-molecular interactions between the two strands of a DNA duplex (see Figure 2.2), you can enclose the border lines of a submatrix with a black box by entering the 0-based indices of the rows and columns within this submatrix in the following format:

- `(starting row, starting column), (ending row, ending column)`

For example, the top-right 3×3 corner of a 6×6 matrix includes

- rows 0, 1, and 2 (`starting row = 0, ending row = 2`);
- columns 3, 4, and 5 (`starting column = 3, ending column = 5`).

Thus, in this field you must enter `(0, 3), (2, 5)` to enclose the top-right 3×3 matrix with a black box, as shown in Figure 2.2 for the DNA example.

If left as "None," no submatrix will be highlighted.

⑧ **Show Empty Diagonal Cells on fp-LED Heat Maps** – Check this box to display empty diagonal cells on fp-LED heat maps.

⑨ **Delete Existing Heat Map Directories** – To remove previously generated heat map directories before creating new ones (e.g., to retain only the current run's TIFF files and delete old SVGs), check this box. If unchecked, only files with the same extension will be overwritten; others will remain.

⑩ **Confirmation Box** – Clicking the "Confirm Parameters to Proceed" button opens a confirmation dialog.

- Click "No" to revise the plot parameters.
- Click "Yes" to proceed. This will temporarily disable the "Plot" tab, create a "Run" tab that will display the status of the LEDAW run, and automatically initiate the LEDAW run. Once the run is complete, the "Plot" tab will become active again, allowing you to fine-tune the heat map settings.

5.1.7. Run Tab

The layout of the 'Run' tab is as shown in Figure 5.6, irrespective of the selected settings.

In LEDAW-GUI, the tabs are created in the sequence: "N-body", "Two-body", and "Plot". The "Run" tab is added only after confirming the last tab in this sequence, which depends on the options selected in the "Home" tab.

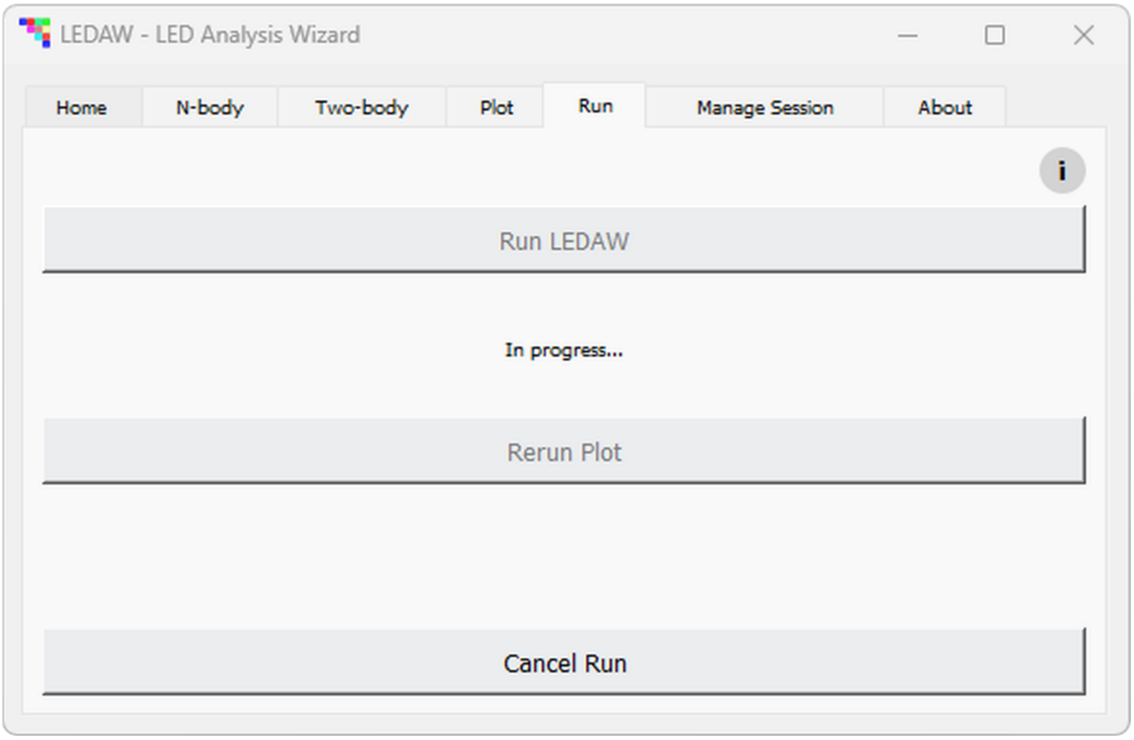


Figure 5.6. The "Run" tab of the LEDAW-GUI.

The "Run" tab displays the status of the LEDAW job and provides controls to cancel or restart the process. Once the tab is created, the LEDAW run starts automatically, and a blinking "*In progress…*" message is shown to indicate that the job is running. When the job finishes, this message changes to "*Completed*".

Depending on the selected options, additional messages may appear, such as:

- *"LED matrices were written to the LEDAW output directory."*
- *"LED matrices and heat maps were written to the LEDAW output directory. Please check the plotted heat maps. If necessary, adjust and confirm the plot parameters in the Plot tab to rerun the plot job."*

If a plot job was requested, the "Plot" tab is reactivated upon completion of the LEDAW run. If the resulting heat maps need adjustments, users should return to the "Plot" tab, revise the parameters as needed, and reconfirm them to rerun the plot generation

The "Rerun Plot" button is enabled only during the initial plot job to prompt the user to modify and confirm plot parameters in the "Plot" tab. After this step, the user is assumed to be familiar with the workflow, and confirming the "Plot" tab alone is sufficient to rerun the job—no need to reactivate the "Rerun Plot" button.

At any point, a running LEDAW job can be cancelled by pressing the "Cancel Run" button. This will display one of the following messages depending on whether the plot job is requested or not:

- *"Cancelled. If needed, press 'Run LEDAW' button to resubmit the job."*
- *"Cancelled. If needed, adjust and then confirm plot parameters in the 'Plot' tab to resubmit the plot job."*

5.2. LEDAW Outputs

5.2.1. LEDAW Output Directory Structure

When a LEDAW job is executed from the "Run" tab, the resulting subdirectory structure within the LEDAW output directory (defined in Field 2 of the "Home" tab, see Figure 5.1) depends on the selected computational settings. This structure is illustrated in Figure 5.7.

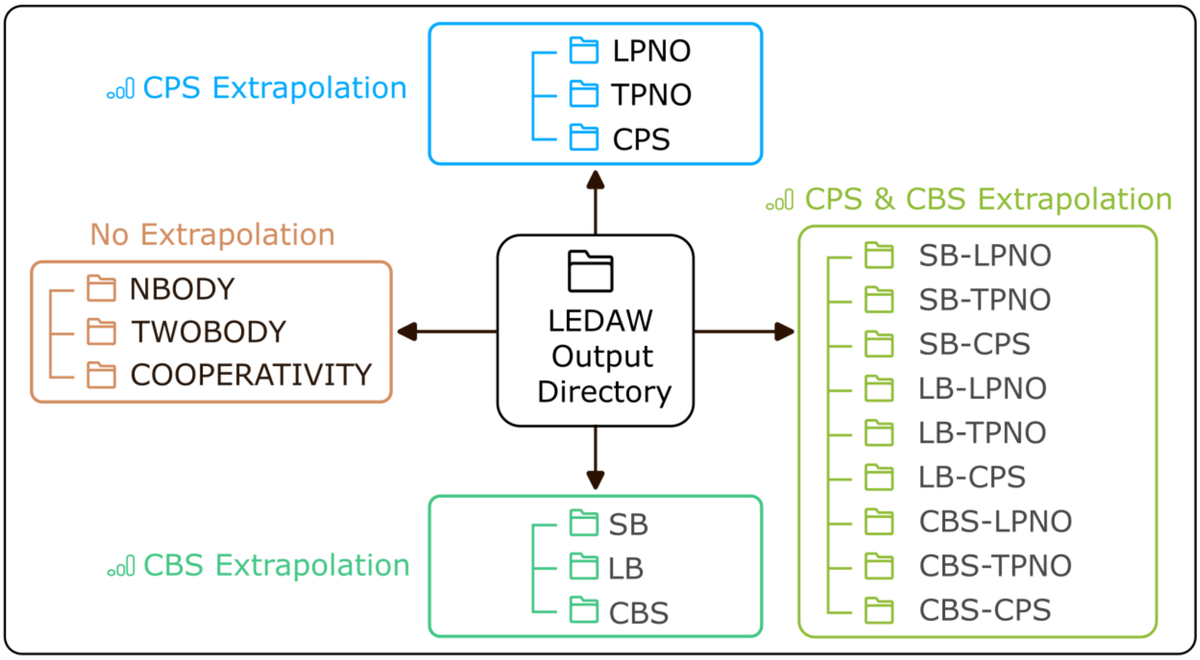


Figure 5.7. Top-level subdirectory layout of the LEDAW output directory, determined by the selected computational settings. When an extrapolation is involved, each subdirectory shown further contains `NBODY`, `TWOBODY`, and/or `COOPERATIVITY` directories.

If N-body LED and/or two-body LED is selected without any extrapolation scheme, the corresponding `NBODY` and/or `TWOBODY` directories are created directly in the LEDAW output directory. If both analyses are selected, a cooperativity analysis is also performed, and the `COOPERATIVITY` directory is likewise created at the top level.

However, if CPS and/or CBS extrapolation is requested, a separate subdirectory is first created for each computational setting (as labeled in Figure 5.7; see Figure 4.1 for abbreviations used in directory names). Within each of these subdirectories, the `NBODY`, `TWOBODY`, and/or `COOPERATIVITY` directories are generated based on the selected analyses.

As shown in Figure 5.8, the `NBODY`, `TWOBODY`, and `COOPERATIVITY` directories each contain two main Excel files ("`Summary_Standard_LED_matrices.xlsx`" and "`Summary_fp-LED_matrices.xlsx`"), which summarize the standard LED and fp-LED interaction energy matrices. If "the Plot" option is selected, two additional directories (`HEAT-MAP-STD-LED` and `HEAT-MAP-STD-LED`) are also generated. These provide heat map versions of the interaction energy matrices from each sheet of the corresponding Excel files.

For users interested in more theory-driven analyses, additional Excel files are available within these directories, offering more detailed LED decompositions (e.g., for strong pairs, weak pairs, triples, and solvent contributions). However, we recommend focusing primarily on the two main summary Excel files listed above, as they present the LED terms in a more intuitive and interpretable format.

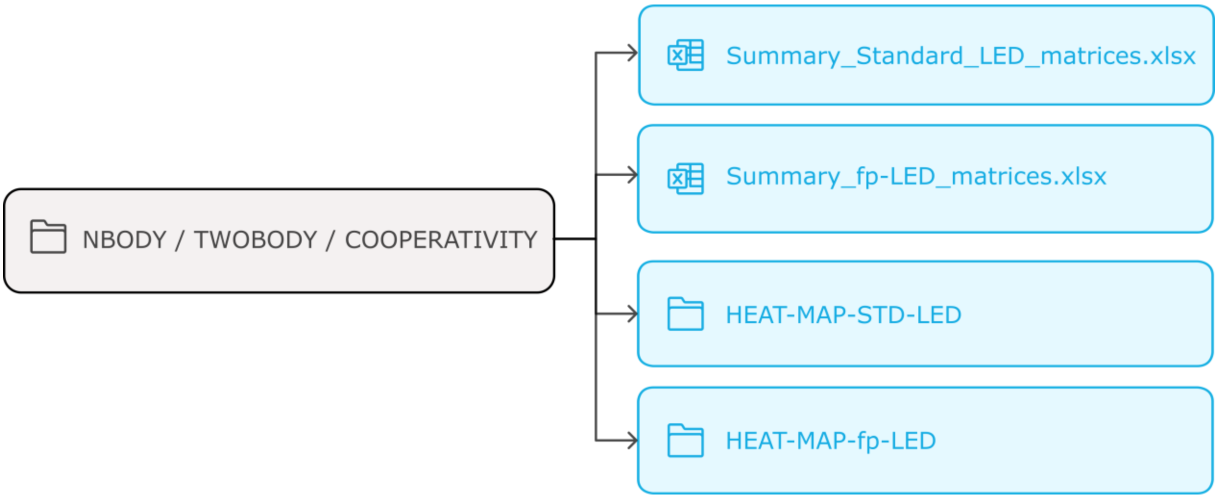


Figure 5.8. Two main Excel files in each of the `NBODY`, `TWOBODY`, and `COOPERATIVITY` LEDAW output directories, containing the LED interaction energy matrices. If heat map plotting is requested, two additional directories are generated, each including heat maps corresponding to the matrices in the sheets of these Excel files.

5.2.2. LED Terms in LEDAW Summary Files

Below, we describe how the total interaction energy (`TOTAL`) is decomposed into LED terms in the summary files shown in Figure 5.8. Figure 5.9 presents LED of DLPNO-CCSD(T) and HFLD interaction energies, with term names as used in LEDAW. For DLPNO-CCSD, the same term names as for DLPNO-CCSD(T) are used, but without the (T) suffix.

DLPNO-CCSD(T), DLPNO-CCSD, and HFLD interaction energies each consist of three components: the reference energy (`REF`), the correlation energy (`C-CCSD(T), C-CCSD, and Disp HFLD`, respectively), and, if an implicit solvation model is applied, a solvent contribution (`SOLV`). In LEDAW, the decomposed `SOLV` term represents the total implicit solute-solvent interaction contributions. This includes:

- The reference-level dielectric interaction energy (`REF-DIEL`);
- The correlation correction to the dielectric energy (`CORR-DIEL`);
- The nonelectrostatic CDS correction (`REF-CDS`).

As the reference part is common among these methods, their `REF` LED terms (based on RHF or QROs) are identical: Electrostatic (`Electrostat`), `Exchange`, and the reference component of the electronic preparation energy (`REF-EL-PREP`).

In the correlation part,

- HFLD includes only the dispersion energy (`Disp HFLD`);
- DLPNO-CCSD(T) and DLPNO-CCSD include:
    - Dispersion energy (`Disp-CCSD(T) or Disp-CCSD`);
    - Nondispersive inter-fragment correlation energy (`Inter-NonDisp-C-CCSD(T)` or `Inter-NonDisp-C-CCSD`);
    - Correlation component of the electronic preparation energy (`C-CCSD(T)-EL-PREP` or `C-CCSD-EL-PREP`).

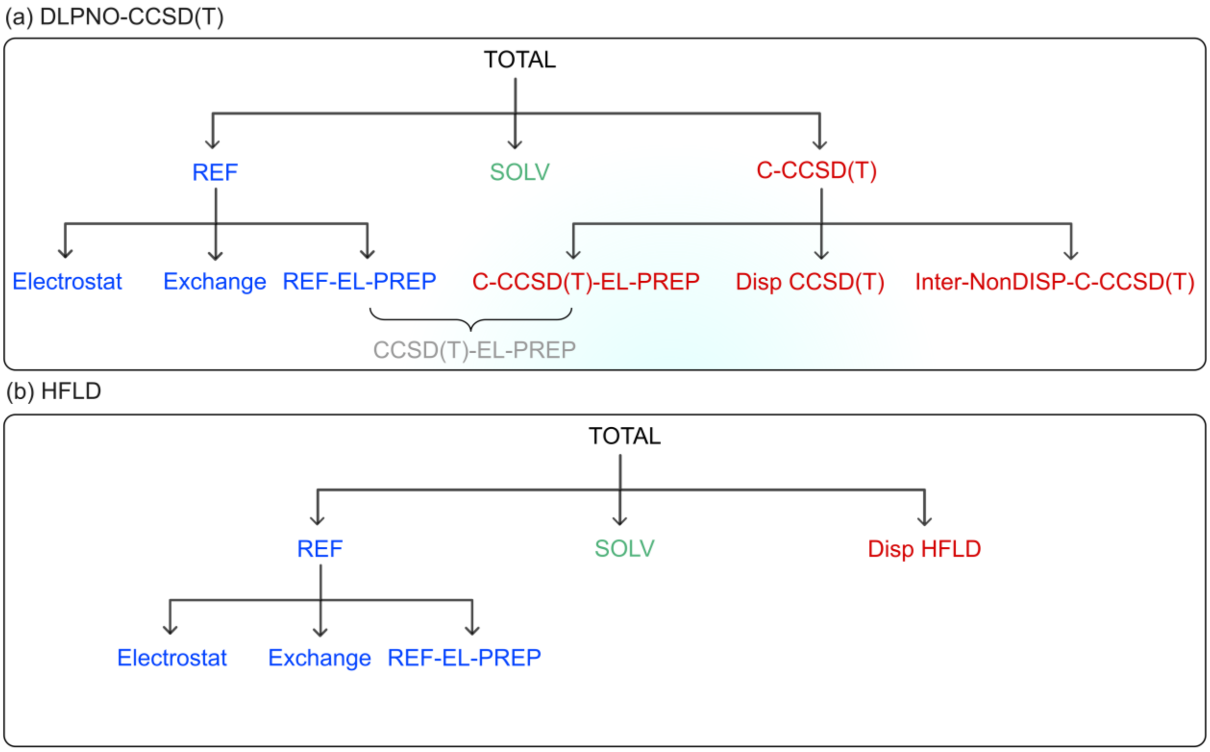


Figure 5.9. LED of (a) DLPNO-CCSD(T) and (b) HFLD interaction energies, with abbreviation of the terms as used in LEDAW output files. The DIEL term appears only when an implicit solvation scheme is applied in the ORCA calculations. For DLPNO-CCSD, the (T) suffix is dropped from the LED term names used in DLPNO-CCSD(T).

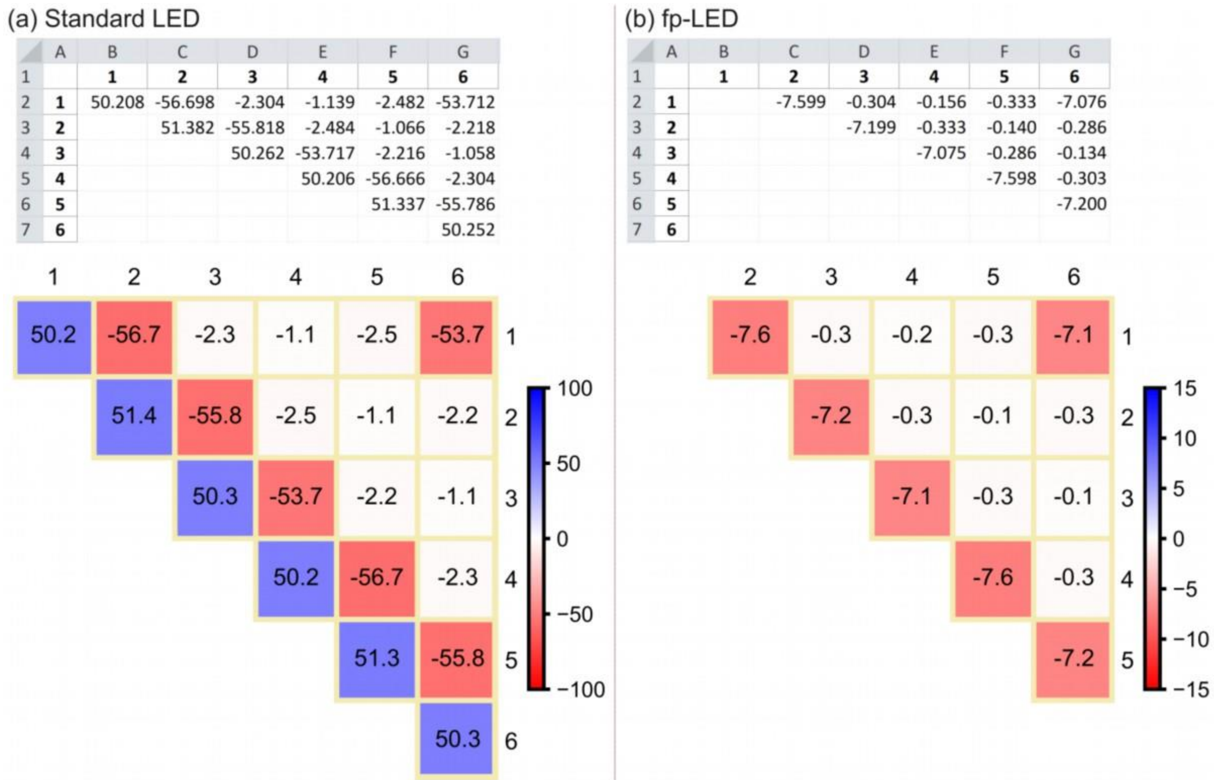
For simplicity in presenting LED results, the reference and correlation components of the electronic preparation energy may also be combined into a single term: `CCSD(T)-EL-PREP` or `CCSD-EL-PREP`. By contrast, HFLD includes only the `REF-EL-PREP` term.

In the standard LED scheme, the reference and correlation electronic preparation energy terms are already included as diagonal elements within the `REF` and `C-CCSD(T)` (or `C-CCSD`) matrices. Therefore, separate Excel sheets and heat maps are not provided for electronic preparation energy within this scheme. However, in the fp-LED scheme, where these contributions appear as non-diagonal terms, they are included in separate sheets and figure files.

5.2.3. Appearance of LEDAW Output Files

As an illustrative example of the appearance of the Excel sheets for matrices and of figure files for corresponding heat maps, Figure 5.10 shows unedited content from the LEDAW output directories for the total DLPNO-CCSD(T_1)/CBS(3/4)/CPS(6/7) interaction energy (`TOTAL`) decomposition of the boat conformer of the water hexamer. The results were generated using relabel mapping applied to the original fragment labels from the ORCA output files, as described in Figure 5.3. The corresponding ORCA output files are available in the LEDAW repository.

As described in Section 5.1.6, the color scale ranges, cell sizes, and resolution of the heat maps are adjustable. A submatrix can also be highlighted by enclosing it in a black box, as shown in Figure 2.2. For fp-LED, diagonal cells can optionally be shown as empty elements for consistency with the standard LED format. Other stylistic adjustments can be easily made if the figure format is chosen in the Plot tab as a vector format (*e.g.*, SVG).



(a) Standard LED

	A	B	C	D	E	F	G
1		1	2	3	4	5	6
2	1	50.208	-56.698	-2.304	-1.139	-2.482	-53.712
3	2		51.382	-55.818	-2.484	-1.066	-2.218
4	3			50.262	-53.717	-2.216	-1.058
5	4				50.206	-56.666	-2.304
6	5					51.337	-55.786
7	6						50.252

(b) fp-LED

	A	B	C	D	E	F	G
1		1	2	3	4	5	6
2	1		-7.599	-0.304	-0.156	-0.333	-7.076
3	2			-7.199	-0.333	-0.140	-0.286
4	3				-7.075	-0.286	-0.134
5	4					-7.598	-0.303
6	5						-7.200
7	6						

Figure 5.10. Unedited Excel sheet (top) and figure file (bottom) contents from the LEDAW output directories for the total DLPNO-CCSD(T_1)/CBS(3/4)/CPS(6/7) interaction energy (`TOTAL` in kcal/mol) decomposition of the boat conformer of the water hexamer within (a) standard LED and (b) fp-LED schemes. Relabel mapping was applied as shown in Figure 5.3.

6

HOW TO CITE

LEDAW Software

1. Altun, A.; Neese, F.; Bistoni, G. LEDAW: An Integrated Software Suite with GUI for Automating Local Energy Decomposition Analysis of Molecular Interactions, *J. Chem. Inf. Model.* **2025**, *65*, 8917−8923.
2. https://github.com/ahmetaltunfatih/LEDAW

LEDAW is built on several previously published methods, which should be cited separately where appropriate. For convenience, key references are grouped below by topic.

LED Methodology

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CBS Extrapolation

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